02 Electronic Displacement Effect

Introduction of GOC and Intermediates:

Bond Fission:

Any organic reaction cannot be completed without breaking or formation of a covalent bond. There are two types of bond breaking :-

(a) Homolytic bond breaking :

In such cases, bonding e^- are equally distributed between bonding atom so that electrically neutral intermediates are formed.



(b) Heterolytic bond breaking :

In such cases, bonding e^- is transferred to any one of the atoms such that charged intermediates are formed.



Note :

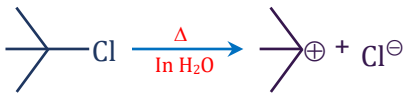
(i) Any covalent bond can break in both homolytic and heterolytic manner.

(ii) Heterolytic bond breaking takes place in polar solvent.

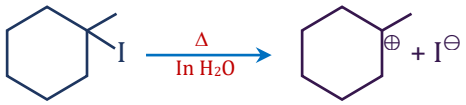
(iii) Homolytic bond breaking takes place in gas phase or in non-polar solvent.

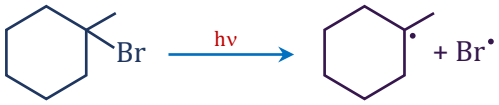
(iv) **Bond energy :** Minimum energy required to break a bond in homolytic cleavage.

$$\text{Bond Energy} \propto \frac{1}{\text{Stability of free radical obtained}}$$

Ex.I (i) 

(ii) 

(iii) 

(iv) 

(v) $Cl_2 \xrightarrow{h\nu} Cl^{\bullet} + Cl^{\bullet}$

(vi) $H—Cl \xrightarrow{H_2O} H^{\oplus}_{(aq)} + Cl^{\ominus}_{(aq)}$

(vii) $\ddot{N}H_3 + H^{\oplus}_{(aq.)} \longrightarrow NH^{\oplus}_{4(aq.)}$

(viii) $CH_4 + H^{\oplus}_{(aq.)} \longrightarrow$ No reaction

(ix) $H—Cl \xrightarrow[\Delta]{gas} H^{\oplus}_{(g)} + Cl^{\ominus}_{(g)}$ (Radicals are formed, then on prolonged heating ions are formed)

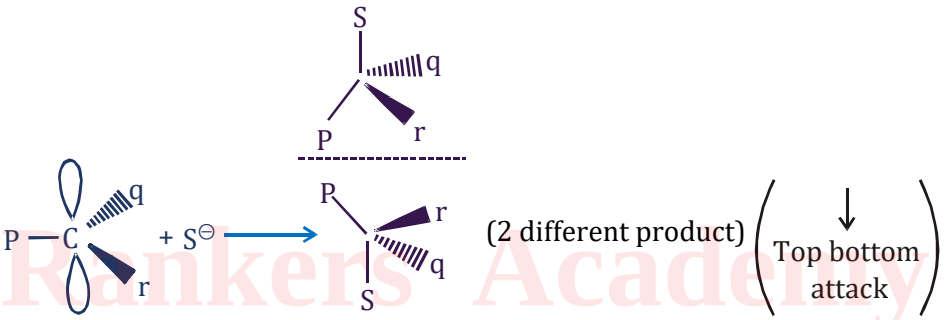
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Intermediates:

Species obtained between reactant and product during reaction.

(Carbocation) $^{\oplus}CH_3$	Carbanion $^{\ominus}CH_3$	Carbon free radical $\dot{C}H_3$
Highly unstable	Highly unstable	Highly unstable
Incomplete octet	Complete octet	Incomplete octet
Only $6e^{\ominus}$	$8e^{\ominus}$	$7e^{\ominus}$
Lewis Acid	Lewis Base	Neither
Bond Pair : 3	Bond Pair :3	Bond Pair : 3
Unpaird Pair (n) : 0	Unpaired $e^{\ominus}$: 0	Unpaired $e^{\ominus}$: 1
Lone Pair : 0	Lone Pair : 1	Lone Pair = 0
Magn. Moment : $\sqrt{n(n+2)}=0$	Mag. moment = 0	Mag. moment = $\sqrt{3}$
Spin Multiplicity = $2\|s\| + 1 = 1$	Spin Multiplicity = 1	Spin Multiplicity = 2
Hyb. = sp^2	Hyb. = sp^3	Hyb. = sp^2
Bond angle = 120°	Bond Angle = 107°	Bond Angle = 108°



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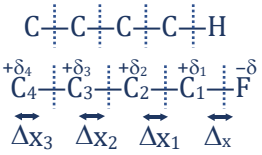
Inductive effect (Groups showing +I and –I effects):

Electron Displacement Effect : (EDE)

Effect because of displacement of electron is known as EDE.

Permanent Effect	**Emporary Effect**
(i) Inductive Effect (displacement of σ-$e^{\ominus}$)	(i) Inductomeric Effect displacement of σ-$e^{\ominus}$
(ii) Resonance/Mesomeric effect (displacement of p(π)$e^{\ominus}$)	(ii) Electromeric Effect temp. displacement of p-$e^{\ominus}$
(iii) Hyperconjugation effect (displacement of σ $e^{\ominus}$ in p-orbital)	

Inductive effect and Stability of Intermediates :



(i) $\Delta x > \Delta x_1 > \Delta x_2 > \Delta x_3$

(ii) $\delta = (\delta_1 + \delta_2 + \delta_3 + \delta_4)$

(iii) $\delta > \delta_1 > \delta_2 > \delta_3 > \delta_4$

(iv) $\delta_4 \sim 0$ (experimentally)

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Type of inductive effect :

–I-effect : (i) Permanent displacement of σ-e$^{\ominus}$ in a covalent bond because of any atom or group is known as I-effect of that atom. If the electron withdrawing power of group is higher than that of hydrogen, then these groups are known as –I groups.

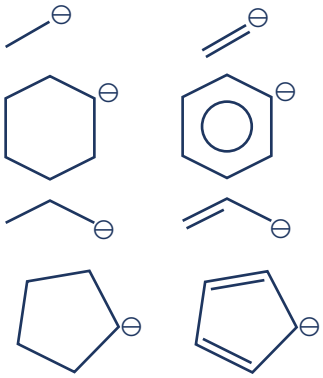
Decreasing order of e$^-$ withdrawing tendency by inducting.

–I-Series :

$-\overset{\oplus}{N}F_3 > \overset{\oplus}{N}R_3 > -\overset{\oplus}{N}H_3 > -NO_2 > -CN > -SO_3H > -CHO > -COOH > -F > -Cl > -Br > -I > -OH > -NH_2 > -C \equiv CH > -Ph > -CH = CH_2 > (H)$

Hybridisation state of carbanion :

* sp^3 sp^2



+I effect :

If the electron donating power of group is higher than that of hydrogen, then these groups are known as +I groups.

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+I series :

$-CH_2^{\ominus} > -NH^{\ominus} > -O^{\ominus} > -COO^{\ominus} > -\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3 > -CH\begin{matrix}CH_3\\CH_3\end{matrix} > -CH_2-CH_3 > -CT_3 > -CD_3 > -CH_3 > -T > -D > -H$

Note:

(i) For I-effect C–H bond is taken as reference.

(ii) It is permanent and weak effect.

(iii) During I-effect partial displacement takes place within orbitals.

(iv) I-effect is distance dependent effect.

(v) Practically, it is observed only up to C_3 atom.

(vi) Experimentally proved, that 10% effect is transferred to next atom.

Ex.1 $\overset{+\delta_1}{CH_2}-\overset{+\delta_2}{CH_2}-\overset{+\delta_3}{CH_2}-\overset{+\delta_4}{CH_2}$ with $F^{-\delta_5}$ and $Cl^{-\delta_6}$

I-effect of Cl: $C_1 - C_4$ ↑ $\quad C_4 - C_1$ ↑

I-effect of F: ↓ ↓

Magnitude : $\delta_5 > \delta_6 > \delta_1 > \delta_4 > \delta_2 > \delta_3$

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Ex.2 **(a)** –T, –D, –H (+I)

Heavier the molecule (atom), less is its oscillation frequency ⇒ Higher overlapping makes strong and small bond. So that as small displacement of e⁻ cause large I-effect. So

$\boxed{-T > -D > -H}$

(b) $-CH_3, -CH_2D, -CHD_2, -CD_3$ (+I)

$\boxed{-CH_3 < -CH_2D < -CHD_2 < -CD_3}$

(c) $-CH_3, -CD_3, -CT_3$ (+I)

$\boxed{-CH_3 < -CD_3 < -CT_3}$

(d) $-CH_3, -SiH_3$(+I)

$-CH_3 < -SiH_3$

(e) $-CH_3, -CCl_3$ (+I)

$-CH_3 > -CCl_3$

Application of I-effect

(1) Stability of carbocation:

(a) If number of + I groups increases then stability of carbocation increases.

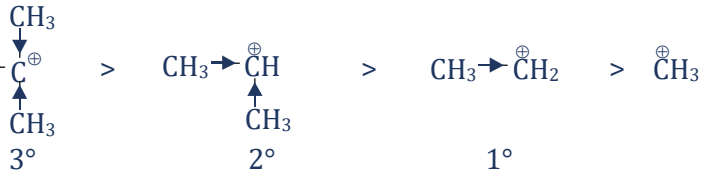
(b) If number of –I groups decreases then stability of carbocation increases.

Therefore

$$\text{Energy} \propto \frac{1}{\text{Stability}}$$

$\boxed{\text{Stability of Carbocation} \propto +\text{I effect}}$

$$\text{Stability of Carbocation} \propto \frac{1}{\text{-I effect}}$$

Ex : Stability order : 

Sol. Reason : More no. of +I group.

more stable carbocation.

(2) Stability of carbanion :

(a) If number of – I groups increases then stability of carbanion increases.

(b) If number of +I groups decreases then stability of carbanion increases.

Therefore

$\boxed{\text{Stability of Carbanion} \propto -\text{I effect}}$

$\boxed{\text{Stability of Carbanion} \propto \frac{1}{+\text{I effect}}}$

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Example:

(1) $CH_3-\overset{CH_3}{\underset{CH_3}{\overset{|}{\underset{|}{C^{\ominus}}}}}$ (2) $\overset{\ominus}{C}H-CH_3$ (3) $CH_3-\overset{\ominus}{C}H_3$ (4) $\overset{\ominus}{C}H_3$

3° 2° 1°

More No. of +I group.
Less stable carbanion.
So stability order 1 < 2 < 3 < 4

Example :

(1) $\overset{\ominus}{C}H_2-\underset{F}{CH}-CH_2-CH_3$ (2) $\overset{\ominus}{C}H_2-CH_2-\underset{F}{CH}-CH_3$ (3) $\overset{\ominus}{C}H_2-CH_2-CH_2-CH_2-F$

Minimum distance of –F.
Maximum –I of –F.
Minimum negative charge.
Maximum stable.
So, stability order 1 > 2 > 3

Acidic and Basic Strength

Acidic strength :



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(i) If No. of – I groups increases then acidic strength increases.
(ii) If No. of + I groups increases then acidic strength decreases.

$$\text{acidic strength} \propto -\text{I effect} \propto \frac{1}{+\text{I effect}}$$

Example :

(i) $CH_3-\underset{\underset{O}{\|}}{C}-OH \xrightarrow{-H^{\oplus}} CH_3-\underset{\underset{O}{\|}}{C}-O^{\ominus}$

+I of $-CH_3$
So anion is less stable

$Cl-CH_2-\underset{\underset{O}{\|}}{C}-OH \xrightarrow{-H^{\oplus}} Cl-CH_2-\underset{\underset{O}{\|}}{C}-O^{\ominus}$

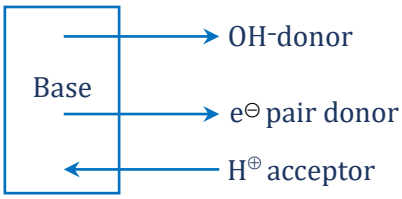
–I of –Cl, so anion is more stable
and corresponding acid is more acidic.

(ii) $CH_3-CH_2-\underset{F}{CH}-COOH > CH_3-\underset{F}{CH}-CH_2-COOH > CH_2-\underset{F}{CH_2}-CH_2-COOH$

minimum distance of F
from –COOH maximum –I
of F. So maximum acidic.

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Basic strength :

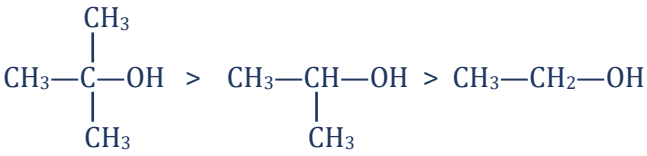


(i) If No. of + I groups increases then basic strength increases.

(ii) If No. of – I groups increases then basic strength decreases.

$$\text{Basic strength} \propto +\text{I effect} \propto \frac{1}{-\text{I effect}}$$

Example :



Maximum +I.

Maximum tendency to donate l.p.

Maximum basic.

Illustration 1

Ethyl amine is more basic then aniline, why?

Solution:

Due to the + I effect of ethyl group.

Illustration 2

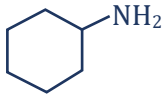
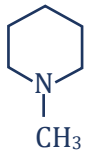
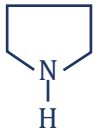
Cl–NH_2 is less basic then methyl amine, why?

Solution:

Due to – I effect of –Cl group and p_π–d_π conjugation.

Illustration 3

Correct basic order of the following different amine in chlorobenzene medium is -

(A) CH_3–CH_2–NH_2 (B)  (C)  (D) 

Solution:

(C > D > B > A)

Illustration 4

Which is most basic among the following :

(A) CH_3NH_2 (B) $CH_3CH_2NH_2$ (C) NH_3 (D) $(CH_3)_2CHNH_2$

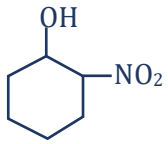
Ans.

(D)

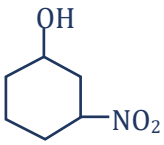
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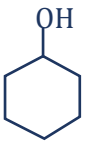
Illustration 5

Which is most acidic compound –

(A) 

(B) 

(C) 

(D) 

Ans.

(A)

(a) Strength of aliphatic carboxylic acids

Example:

(I) $H_3C—H_2C—H_2C—\underset{\underset{Cl}{|}}{CH}—COOH$

(II) $CH_3—CH_2—\underset{\underset{Cl}{|}}{CH}—CH_2—COOH$

(III) $CH_3—\underset{\underset{Cl}{|}}{CH}—CH_2—CH_2—COOH$

(IV) $\underset{\underset{Cl}{|}}{CH_2}—CH_2—CH_2—CH_2—COOH$

Acid strength order is : I > II > III > IV

Explanation : We know that – I effect increases acidic strength. But it is distance dependent effect so where the – I group is nearest to – COOH, it exerts strong effect and makes acid stronger.

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(b) Acidic strength of alcohol : Greater - I effect results in greater acidic character.

Similarly, greater + I effect lowers acidic character.

Example : Acidic strength order is :

$$CH_3 - OH > CH_3 \rightarrow CH_2 - OH > \begin{matrix} CH_3 \\ CH_3 \end{matrix} \!\!\searrow\!\!\nearrow CH-OH > \begin{matrix} CH_3 \\ CH_3 \\ CH_3 \end{matrix} \!\!\searrow\!\!\rightarrow\!\!\nearrow C—OH$$

Illustration 6

Find out the strongest acid of the following :

(A) $O_2N–CH_2–COOH$

(B) $F–CH_2–COOH$

(C) $H_3CO–CH_2–COOH$

(D) $CH_3–CH_2–COOH$

Solution:

(A) Since, $–NO_2$ has strong – I effect its influence will make corresponding acid strongest.

Illustration 7

Which is most basic among the following :

(A) CH_3NH_2 (B) $CH_3CH_2NH_2$ (C) NH_3 (D) $(CH_3)_2CHNH_2$

Solution:

(D) Isopropyl group has maximum +I effect in the given alkyl group so corresponding nitrogen atom is most electron dense.

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Resonance, Conditions for resonance, Stability of Resonating structures:

Resonance Effect :

If a single structure cannot explain all of the properties of a molecule then more than one structures are required then it is said that molecule is showing resonance. It is due to the delocalization of electrons within the molecule.

All the contributing structures in the resonance are called **resonating structures** or **canonical structures.**

These structures are helpful in explanation of chemical reactivity or the chemical reaction of the compound.

(i) Resonating structure are not the real structures.

(ii)The real structure is a hybrid of all resonating structures which is called as **resonance hybrid**.

Conditions for resonance:

1. System must be planar.
2. System must be in conjugation (i.e. parallel p orbitals are required)

Following type of conjugation may be present in a molecule :-

1. $\pi \rightarrow \pi$ conjugation :

If there are two π bonds at alternate position, then $e^{\ominus}$ of one π bond are transferred towards another π bond. (According to I–effect).

Example :

(i) $CH_2=CH-CH=CH_2 \longleftrightarrow \overset{\oplus}{C}H_2-CH=CH-\overset{\ominus}{C}H_2 \longleftrightarrow \overset{\ominus}{C}H_2-CH=CH-\overset{\oplus}{C}H_2$

(ii) $CH_2=CH-\underset{\underset{O}{\|}}{C}-H \longleftrightarrow \overset{\oplus}{C}H_2-CH=\underset{\underset{O^{\ominus}}{|}}{C}-H$

(iii) $CH_3-CH=CH-CH=CH_2 \longleftrightarrow CH_3-\overset{\oplus}{C}H-CH=CH-\overset{\ominus}{C}H_2$

2. Lone pair $\rightarrow \pi$ conjugation :

If there is one lone pair and one π bond is in parallel orbital, then electrons of lone pair are delocalised through π bond.

Example :

(i) $CH_2=CH-\ddot{O}H \longleftrightarrow \overset{\ominus}{C}H_2-CH=\overset{\oplus}{O}H$

(ii) $CH_2=CH-\overset{\ominus}{C}H_2 \longleftrightarrow \overset{\ominus}{C}H_2-CH=CH_2$

(iii) $CH_2=CH-\ddot{O}-CH_3 \longleftrightarrow \overset{\ominus}{C}H_2-CH=\overset{\oplus}{O}-CH_3$

(iv) $CH_2=CH-\overset{\ominus}{O} \longleftrightarrow \overset{\ominus}{C}H_2-CH=O$

3. $\pi \rightarrow$ vacant orbital conjugation :

If there is one vacant orbital and one π bond are in parallel orbital then electrons of π bond are delocalised towards vacant orbital.

Example :

$CH_2=CH-\overset{\oplus}{C}H_2 \longleftrightarrow \overset{\oplus}{C}H_2-CH=CH_2$

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4. $\pi \rightarrow$ unpaired e$^{\ominus}$ conjugation :

If there is one free e$^{\ominus}$ and one π bond are at alternate position.

Example :

(i) $CH_2=CH-\dot{C}H_2 \longleftrightarrow \dot{C}H_2-CH=CH_2$

(ii) $CH_3-CH=CH-\dot{C}H_2 \longleftrightarrow CH_3-CH-\dot{C}H=CH_2$

5. Lone pair $\rightarrow$ +ve charge conjugation :

If there is one lone pair or negative charge and one positive charge are at adjacent atoms then e$^{\ominus}$ of lone pair or negative charge are transferred towards positive charge.

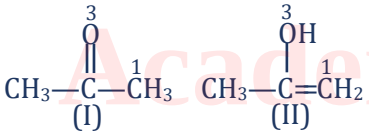
Example : $\overset{\oplus}{C}H_2-\ddot{O}H \longleftrightarrow CH_2=\overset{\oplus}{O}H$

Conditions of Resonating Structures : Resonance structures should fulfil following conditions :

(a) All resonating structures must have the same arrangement of atomic nuclei. Resonance differs from tautomerism in this very important aspect.



Positions of atomic nuclei in (I) and (II) are same.



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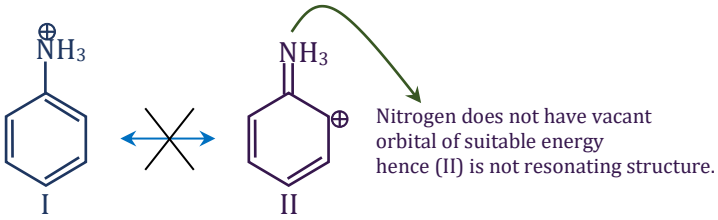
Position of hydrogen nuclei in (I) and (II) are different, hence (I) and (II) are not resonating structures, they are tautomer.

(b) The resonating structures must have the same numbers of paired and unpaired electrons. However, they differ in the way of distribution of electrons.



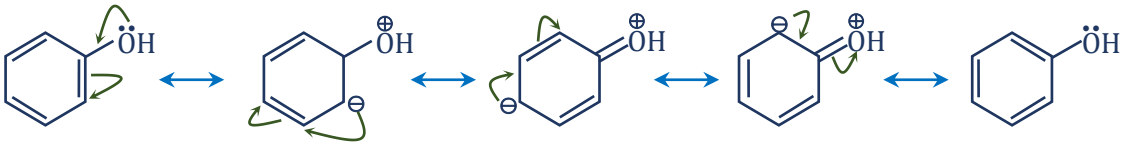
Total number of	Total number of
paired electrons =16	paired electrons =16
unpaired electron = 1	unpaired electron = 1

- The energy of the different resonating structures must be the same or nearly the same.
- All atoms that are part of the delocalisation system must be in a plane or be nearly planar.

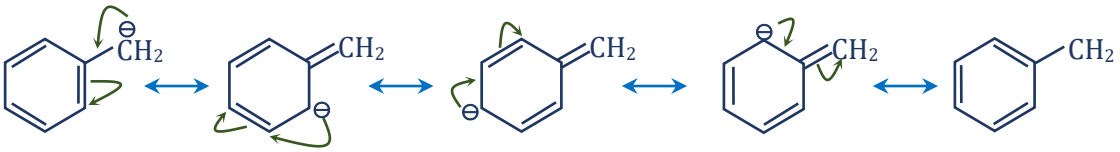


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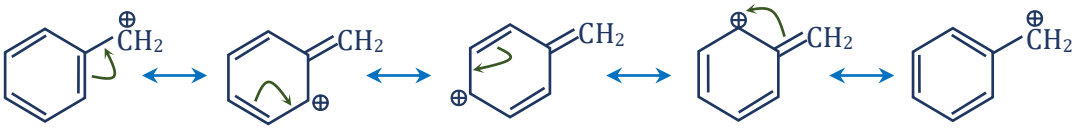
Examples of drawing resonating structures :

1. 

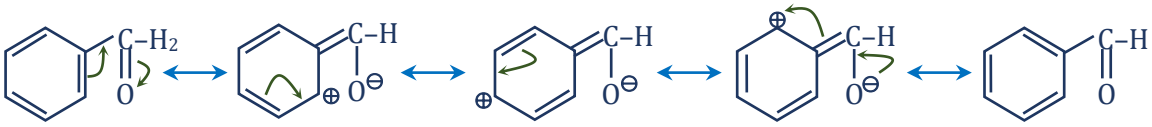
5-Resonating structure

2. 

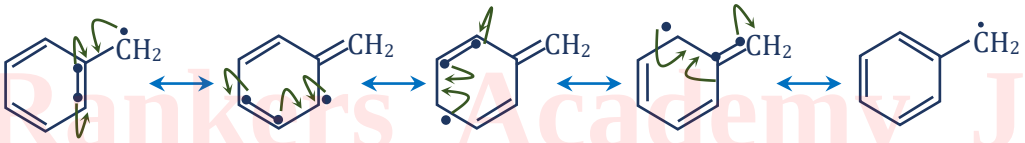
5-Resonating structure

3. 

5-Resonating structure

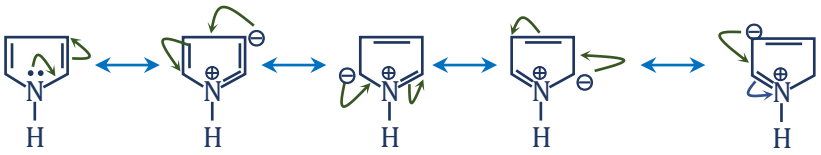
4. 

5-Resonating structure

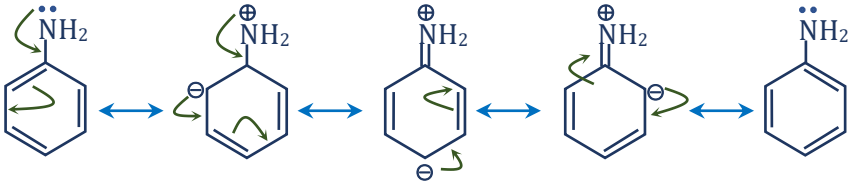
5. 

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5-Resonating structure

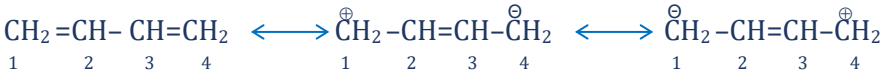
6. 

5-Resonating structure

7. 

5-Resonating structure

Hybrid resonating structure :



(I) (II) (III)

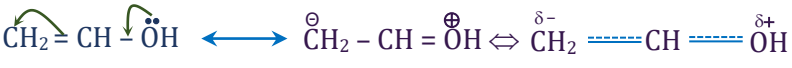
In the structure I C_1–C_2 has double bond while in II and III C_1–C_2 has single bond, so C_1—C_2 shows bond length in between single and double bond similarly in structure I C_2—C_3 has single bond. While in structure II and III C_2—C_3 has double bond, so C_2–C_3 shows bond length in between single and double bond.

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So hybrid resonating structure from all its resonating structures is:

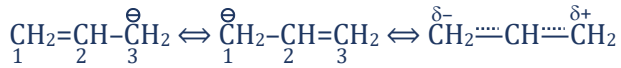
$\Leftrightarrow CH_2 \text{------} CH \text{------} CH \text{------} CH_2$ Hybrid structure

Example :

(i) 

I II

(Resonating structure) (Hybrid structure)

(ii) 

(Resonating structure) (Hybrid structure)

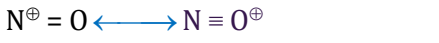
Stability of Resonating structures:

Rules for stability of R.S.:

1. R.S. having complete octet is relatively more stable.

 II > I

I II

 II > I

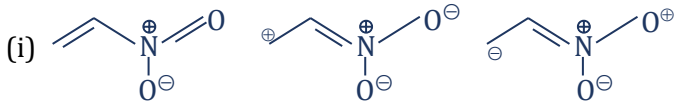
 II > I

2. R.S. having higher no. of covalent bond/(less charge) is more stable.

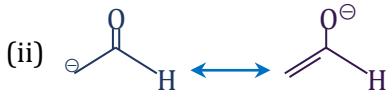


I > II

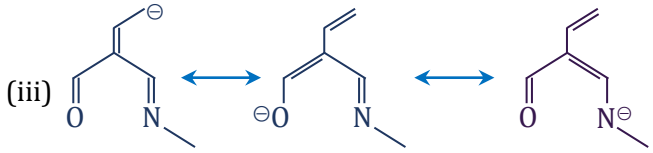
3. R.S. having –ve charge on E.N. atom and +ve charge on Electropositive atom is stable.

(i) 

I > II > III

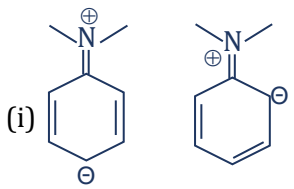
(ii) 

II > I

(iii) 

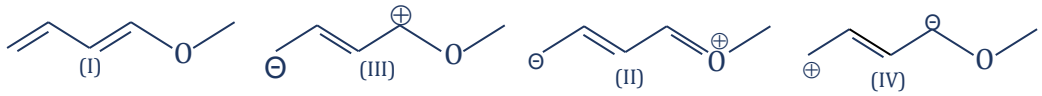
II > III > I

4. R.S. having unlike charges closer and like charges away are relatively more stable.

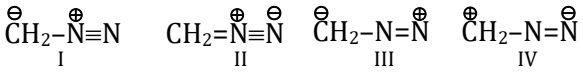
(i) 

II > I

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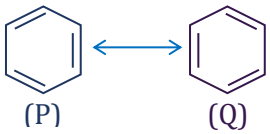
(ii) 

I > II > III > IV

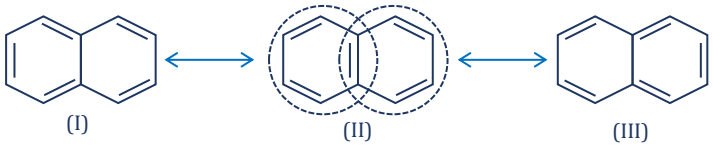
(iii) 

II > I > IV > III

5. R.S. having higher no. of benzenoid structure is more stable.

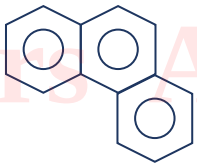
(i) 

I = II

(ii) 

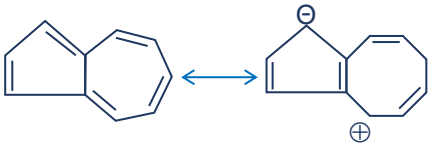
II > I = III

(iii) Anthracene 

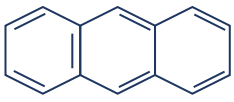
(iv) Phenanthracene 

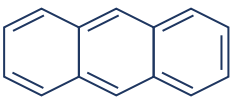
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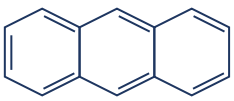
(v) (a) Napthalene is insoluble in aq. acid

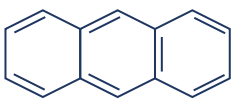
(b) Azulene is soluble in aq. acid, 

6. Anthracene:


(I)


(II)

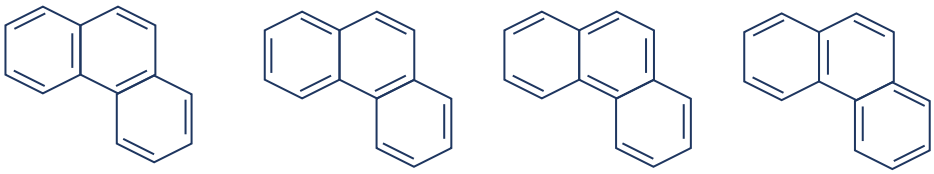

(III)


(IV)

I = IV > II = III

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7. Phenanthrene :



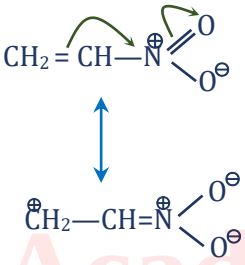
(I) (II) (III) (IV)

I > II = IV > III

Mesomeric effect (Groups showing +M and -M effects):

Mesomeric effect or Resonance effect :

- Polarity developed in conjugated system by interaction of parallel orbitals is known as mesomeric effect.
- If transfer of pi-bond electron takes place from conjugate system to group then it is known as **negative mesomeric (–M)** effect.



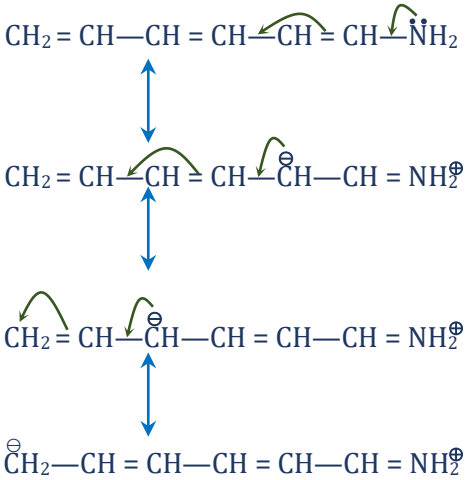
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- For –M effect, group should have either be positive charge or should have vacant orbital.
- Due to –M effect positive charge comes over conjugate system or due to –M effect electron density decrease in conjugate system, such type of conjugate system will be more reactive towards nucleophile or will be less reactive towards electrophile.

Group which shows –M effect are -

$-NO_2$, –CN, $-SO_3H$, – CHO, –COR, –COOH, –COOR, –COX, $-CONH_2$ etc.

If transfer of non-bonding electron takes place from group to conjugate system then it is known as **positive mesomeric (+M)** effect.



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- For +M effect, group should have either be lone pair of electrons or should have negative charge.
- Due to +M effect negative charge comes over conjugate system or electron density increase on conjugate system such type of conjugate system will be more reactive towards electrophile or will be less reactive towards nucleophile.
- **Group which shows + M effect are -**

  $\ddot{O}^{\ominus}$, $\ddot{N}H^{\ominus}$, $-NR_2$, $-NHR$, $-NH_2$, – OH, –OR, –SH – SR, – F, – Cl, – NHCOR, –O–COR etc.
- In mesmeric effect polarity or charge migrate from one end to another end. During charge transfer, energy releases from the conjugate system which increase the stability of conjugate system also.

Aromaticity:

Order of stability for prototype structure

Aromatic > Non-Aromatic > Anti Aromatic

Aromatic :-

Extra stable molecules/ions

→ Cyclic, planar, conjugated,

$(4n + 2)\ \pi e^{\ominus}$ [Huckel's Rule]

n = integer

$2\pi e^{\ominus}$, $6\pi e^{\ominus}$, $10\pi e^{\ominus}$, $14\pi e^{\ominus}$

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Antiaromatic :

→ Extra unstable molecules/ions

Cyclic, planar, conjugated,

$4n\pi e^{\ominus}$

n = integer

$4\pi e^{\ominus}$, $8\pi e^{\ominus}$, $12\pi e^{\ominus}$, $16\pi e^{\ominus}$

	Cyclic	Planar (Nearly)	Conjugated	πe^-
Aromatic 	✓	✓	✓	$6\pi e^{\ominus}$
Antiaromatic 	✓	✓	✓	$4\pi e^{\ominus}$
Non-aromatic 	✓	✓	×	$2\pi e^{\ominus}$

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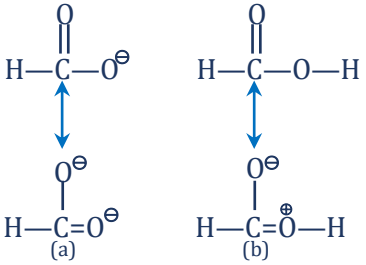
No.	Compound	Cyclic	Planar	Conjugated	$\pi e^{\ominus}$	decision
1.		✓	×	×	$2\pi e^{\ominus}$	Non–aromatic
2.		✓	✓	✓	$2\pi e^{\ominus}$	Aromatic
3.		✓	✓	✓	$4\pi e^{\ominus}$	Anti aromatic
4.		✓	✓	✓	$4\pi e^{\ominus}$	Anti aromatic
5.		✓	✓	✓	$2\pi e^{\ominus}$	Aromatic
6.		✓	✓	✓	$6\pi e^{\ominus}$	Aromatic
7.		✓	×	×	$6\pi e^{\ominus}$	Non–Aromatic
8.		✓	✓	✓	$4\pi e^{\ominus}$	Anti–Aromatic
9.		✓	✓	✓	$6\pi e^{\ominus}$	Aromatic
10.		✓	✓	✓	$6\pi e^{\ominus}$	Aromatic

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Resonance Energy:

Difference in energy between most stable resonating structure resonance hybrid.

Resonance energy ↑ with better overlapping.



(a) has more R.E. than (b)

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(a) has more R.E. than (b)

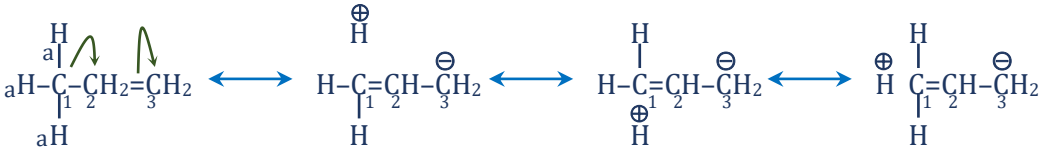
Hyperconjugation:

It is delocalisation of σ electron with p-orbital. Also known as σ –π conjugation or no bond resonance. It may take place in alkene, alkyne, carbocation, free radical, alkyl benzene nucleus. It was given by Nathan and Baker.

Necessary Condition of H effect: Presence of at least one hydrogen at saturated carbon which is α with respect to alkene, alkynes, carbocation, free radical, alkyl benzene nucleus. Carbon which is attached to sp^2 C is called as α –C and H which are attached to α-C are called as α-H.

1. **Hyperconjugation in alkene :**

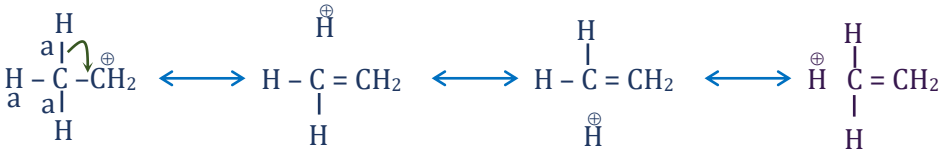
If there is one C–H σ bond and π bond are at alternate position then there will be H-effect.



These resonating structures only suggest that there is some ionic character between C–H bond. Carbon–carbon double bond acquires some single bond character.

2. **Hyperconjugation in carbocation:**

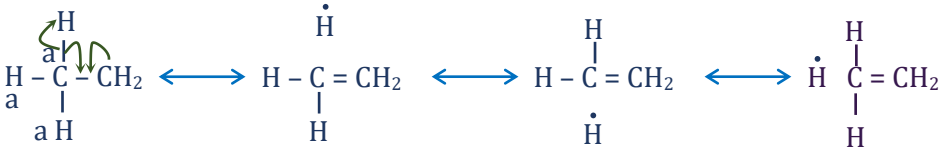
If there is one C–H σ bond and one positive charge are at alternate position, then there will be H–effect. All the structures are called as hyperconjugation structures.



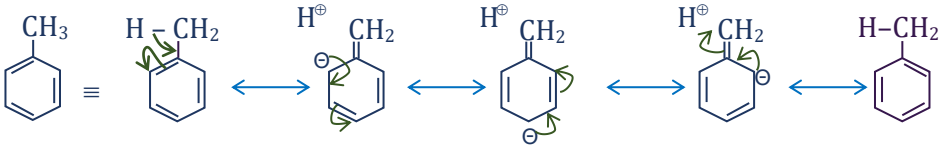
If number of α – H are more then there will be more number of hyperconjugation structures, results more stable carbocation.

3. **Hyperconjugation in free radical :**

If there is one C–H σ bond and one free electron are at alternate position, then there will be H-effect.



4. **Hyperconjugation in toluene :**



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Example :

Structure	Number of α-hydrogens	Total number of Hyper conjugative structure	Number of Hyper conjugative structure (involving C–H bond)
$CH_3–CH=CH_2$	3	4	3
$CH_3–CH_2–CH=CH_2$	2	3	2
$CH_3–CH=CH–CH_3$	6	7	6
$CH_3–\overset{\oplus}{C}H_2$	3	4	3
$CH_3–\overset{\oplus}{C}(CH_3)(CH_3)$	9	10	9

5. Applications of hyper conjugation :

(I) Stability of Alkenes :- More is the number of hyper conjugative structures, more stable is the alkene. So **"More alkylated alkenes are more stable".**

Example : $(H_3C)_2C=C(CH_3)_2$ $\quad$ $(H_3C)_2C=CH-CH_3$ $\quad$ $(H_3C)_2C=CH_2$

Stability in decreasing order

Example : $CH_3–CH=CH_2 > CH_2=CH_2$

3 α-H $\quad$ 0 α-H

more stable

Example : Stability order of alkenes is

$(CH_3)_2C=C(CH_3)_2 > (CH_3)_2C=C(CH_3)H > (CH_3)_2C=CH_2 > H(CH_3)C=C(H)CH_3$

$(CH_3)_2C=C(CH_3)H > (CH_3)H C=C(H)H > H_2C=CH_2$

(II) Bond Length : Bond length is also affected by hyperconjugation

Example: $\overset{H}{\underset{|}{}}H_2\overset{III}{C}-\overset{II}{HC}=\overset{I}{CH_2} \longrightarrow \overset{\oplus}{H}\,H_2\overset{III}{C}=\overset{II}{CH}-\overset{\ominus}{\overset{I}{CH_2}}$

(i) Bond length of C(II) – C(III) bond is less than normal C–C bond.

(ii) Bond length of C(II) – C(I) bond is more than normal C=C bond.

(III) Dipole moment : Since hyperconjugation causes the development of charge, it also affects the dipole moment of the molecule.

Example : $CH_2=CH_2$ < $CH_3–CH=CH_2$ (Dipole moment)

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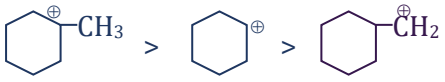
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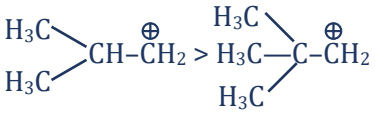
(IV) Stability of carbocation : If hyperconjugation structures are more then more is the stability.

Example: $CH_3-\overset{\oplus}{C}(CH_3)_2 > CH_3-\overset{\oplus}{C}H-CH_3 > CH_3-\overset{\oplus}{C}H_2 > \overset{\oplus}{C}H_3$

9α–H $\quad 6\alpha$–H $\quad 3\alpha$–H $\quad 0\alpha$–H

Maximum stable

Example: 

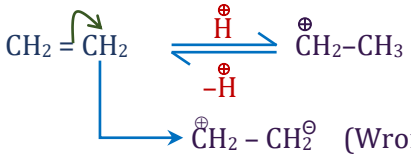
Example: $CH_3-\overset{\oplus}{C}H_2 > CH_3-CH_2-CH_2-\overset{\oplus}{C}H_2 >$  $>$

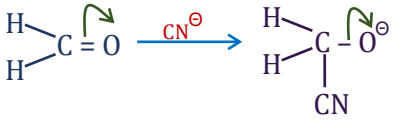
(V) Stability of free radical : Greater the number of α-hydrogen results greater stability of carbon free radical.

Example : $\dot{C}H_3 < CH_3-\dot{C}H_2 < CH_3-\dot{C}H-CH_3 <$ 

Electromeric effect:

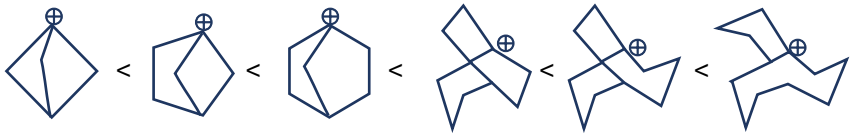
Complete shift of π-electron under the influence of reagent. Temporary effect Operates only in the presence of reagent, vanishesh as reagent removed.

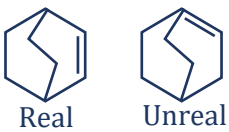
Ex.1 

Ex.2 

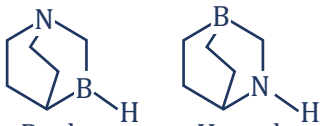
Bredt's Rule:

According to this rule, planarity can't be achieved at bridge head centre of bicyclic compound having less than 8C. (* included). (* p-orbital can't be on bridge head)

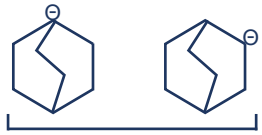
1. 

2. 

Real Unreal

3. 

Real Unreal

4. 

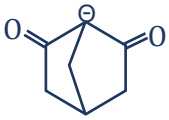
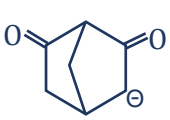
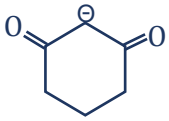
Real

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5.  Real

6.  Unreal  Real

7.  Real  Unreal

8.  (I)  (II)  (III)

Stability (III) > (II) > (I)

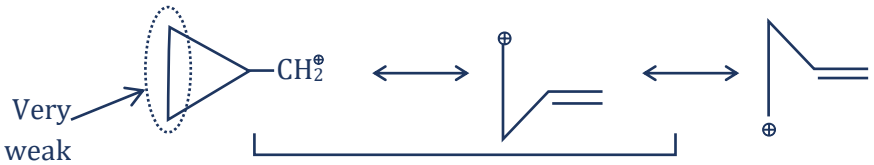
III → Resonance with two

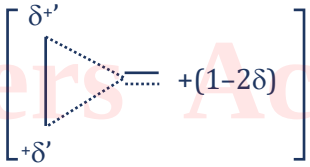
II → Resonance with one

I → No Resonance

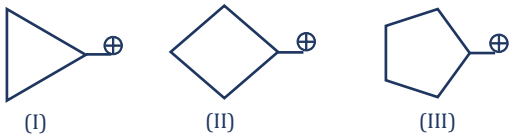
Sigma Resonance

For cyclopropyl methyl carbocation

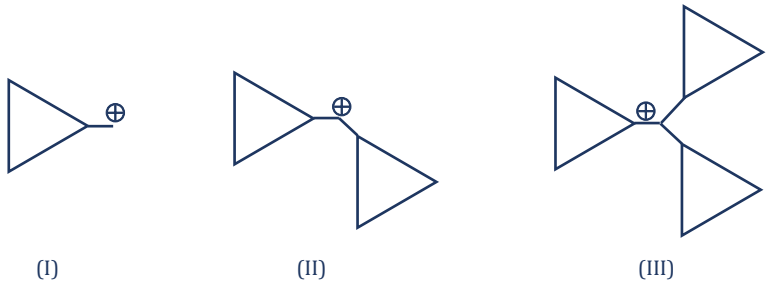




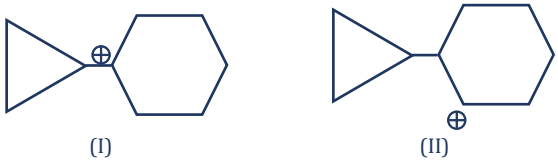
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1.  (I) (II) (III)

I > III > II, 1-σ-resonance, +I effect is higher in III than in II

2.  (I) (II) (III)

III > II > I, σ-resonance increases

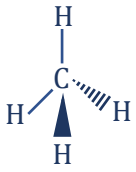
3.  (I) (II)

I > II, σ-resonance possible in I effectively

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Baeyer's Strain Theory:

→ For any sp^3 carbon maximum stability is attained when B.A. is 109°.28'.



According to Baeyer's :



$\delta = \frac{1}{2}[109°28' - \alpha]$

$\alpha = 60°, \delta = 24.5°$

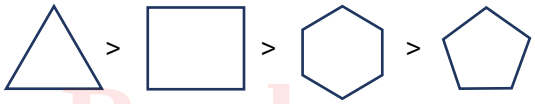
$\alpha = 90°, \delta = 9.5°$

$\alpha = 108°, \delta = 0.5°$

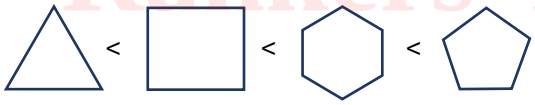
$\alpha = 120°, \delta = -5.5°$

Stability $\propto \frac{1}{\text{strain}}$

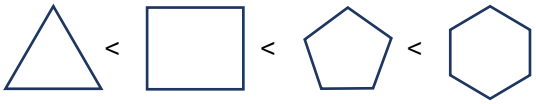
⇒ Strain order :



⇒ Stability order : (Ace. to Baeyer)



Now six-member ring shows flipping which results six-member ring more stable than five-member ring.
overall stability order



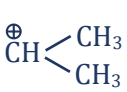
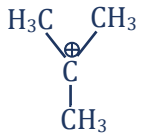
Applications of Electronic displacement effect; Stability of Intermediates and rearrangement of Carbocation:

Stability of Intermediates:

Stability of carbocation ∝ EDG (+R / +H / +I)

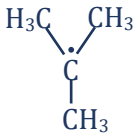
Stability of carbon free radical ∝ EDG (+R / +I)
EWG (–R)

Stability of carbonion ∝ EWG (–R / –I)

1. $+CH_3$, $+CH_2-CH_3$,  , 

(I) (II) (III) (IV)

(I)<(II)<(III)<(IV)

2.  , $-CH_3$,  , 

(I) (II) (III) (IV)

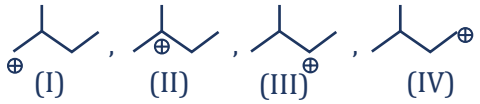
IV > III > II > I

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3. $-CH_3$, $-CH_2-CH_3$, $\overset{\ominus}{C}H(CH_3)_2$, $H_3C-\overset{\ominus}{C}(CH_3)_2$

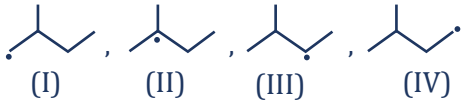
(I) (II) (III) (IV)

I > II > III > IV

4. 

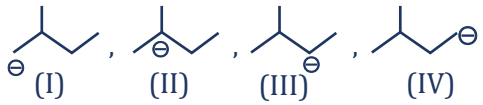
(I) (II) (III) (IV)

II > III > IV > I

5. 

(I) (II) (III) (IV)

IV < I < III < II

6. 

(I) (II) (III) (IV)

IV > I > III > II

7. $CH_3-\overset{\oplus}{C}H_2$, $CH_2=\overset{\oplus}{C}H$, $CH\equiv C^{\oplus}$

(I) (II) (III)

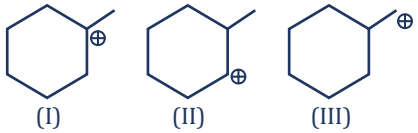
I > II > III

8. $\underset{(I)}{CH_3}-\overset{\bullet}{\underset{(II)}{C}}H_2$, $CH_2=\overset{\bullet}{\underset{(III)}{C}}H$, $CH\equiv\overset{\bullet}{C}$

I > II > III

9. $\underset{(I)}{CH_3}-\overset{\ominus}{C}H_2$, $CH_2=\overset{\ominus}{\underset{(II)}{C}}H$, $CH\equiv\overset{\ominus}{\underset{(III)}{C}}$

III > II > I

10. 

(I) (II) (III)

I > II > III

Heat of hydrogenation:

Heat of hydrogenation : Amount of heat released when one mole of H_2 is added to unsaturated hydrocarbon. Greater the number of α hydrogen results in greater stability of alkene. Thus, greater extent of hyper conjugation results lower value of heat of hydrogenation. So, order for heat of hydrogenation of some alkene can be given as

$CH_2=CH_2 > CH_3-CH=CH_2 > CH_3-CH=CH-CH_3$

Heat of Combustion:

Enthalpy change when 1 mole of compound is oxidized.

$$HOC \propto \text{No. of C-atom} \propto \frac{1}{\text{stability}} \propto \text{strain}$$

Arrange in decreasing order of HOC

1. (I)  (II)  (III)  (IV) 

IV > III > II > I, ↑ no. of C-atom

2. (I)  (II) 

I > II, II is more stable due to +H effect

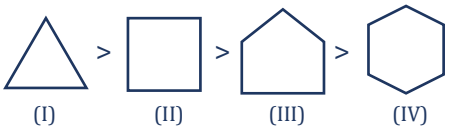
3. (I)  (II) 

I > II, II is more stable due to +H effect

4. (I)  (II)  (III)  (IV) 

IV > III > II > I, increase in no. of carbon.

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5. Per CH_2 in  (I) (II) (III) (IV)

I > II > III > IV the strain decreases from I → IV

6. (I)  (II) 

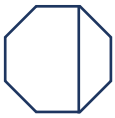
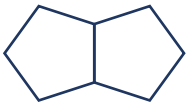
II > I, more no. of C, more is H.O.C.

7. (I)  and (II) 

II > I more no of c, more is H.O.C

8. (I)  (II)  (III) 

III > I > II, no. of C-atom increases increasing H.O.C.

9. (I)  (II)  (III) 

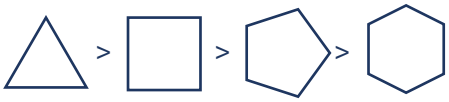
I > II > III, segment attached ring of 3, 4 and 5

10. (I)  (II)  (III) 

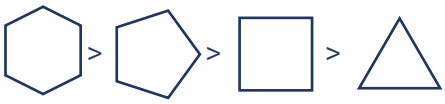
I > II > III, by stability

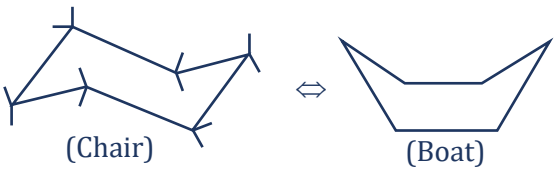
11. From HOC data,

strain order :



Stability order :



 (Chair) ⇔ (Boat)

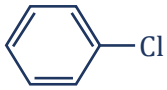
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Bond length and Bond order:

Applications :

(1) Comparison of bond length C–Cl bond length

(a) $CH_2 = CH—\ddot{Cl}:$ (b) $CH_3 –CH_2 –CH_2 –Cl$ (c) 

Ans. (b) > (a) > (c) (C – Cl bond length)

(2) Compare C = C bond length

(a) $CH_2 = CH_2$ (b) $CH_2 = CH_2 – NH_2$

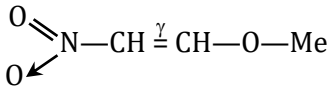
Ans. (b) > (a)

Compare bond length of α, β and γ

$CH_2 \overset{\alpha}{=} CH_2$ $CH_2 = CH \overset{\beta}{—} CH = CH_2$ $CH_3 \overset{\gamma}{—} CH_3$

$\boxed{\gamma > \beta > \alpha}$

Compare bond length in α, β and γ

(a) $CH_2 \overset{\alpha}{=} CH_2$ (b) $CH_2 \overset{\beta}{=} CH—O—Me$ (c) 

$\boxed{\gamma > \beta > \alpha}$

amount of energy required to rotate the bond (it. energy barrier) decreases with the increase in bond length or single bond character.

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$$\text{Bond order} \propto \frac{1}{\text{Bond length}}$$

$$\propto \frac{1}{\text{single bond character}}$$

$\propto$ Bond energy

$\propto$ Bond Stability

Bond Dissociation Energy:

$$\text{B.D.E.} \propto \text{Bond order} \propto \frac{1}{\text{stability of radical formed after bond breaking}}.$$

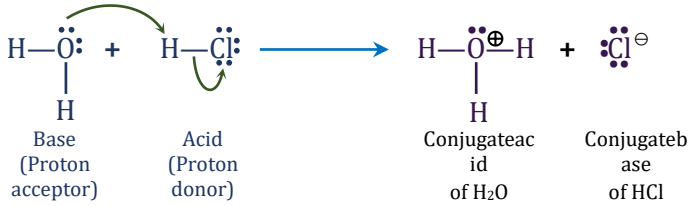
Acidic strength:

It can be explained by Bronsted -Lowry of Lewis concept.

(1) The Bronsted - Lowry concept of Acids and Bases

According to it - an acid is a substance that can donate a proton, and a base is a substance that can accept a proton.

Consider, that gaseous hydrogen chloride dissolves in water :



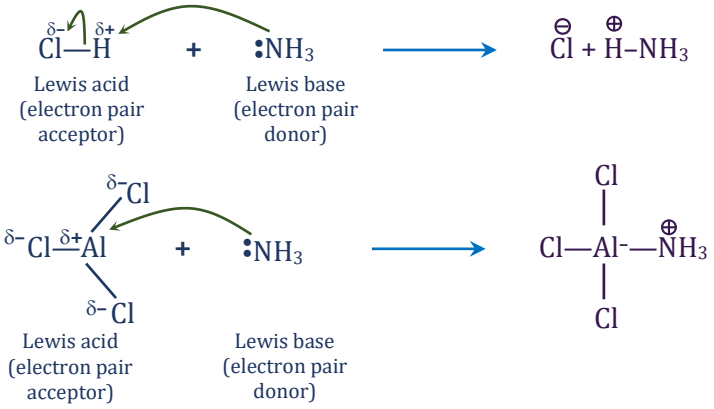
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Hydrogen chloride, a very strong acid, transfer its proton to water. Water acts as a base and accept the proton. The products that result from this reaction are hydronium ion ($H_3O^{\oplus}$) and chloride ion ($Cl^{\ominus}$).

The molecule or ion that forms when an acid donates its proton is called the **conjugate base** of that acid. (The chloride ion is the conjugate base of HCl). The molecule or ion that is formed when a base accepts a proton is called the **conjugate acid** of that base ($H_3O^{\oplus}$ is conjugate acid of H_2O).

(2) The Lewis concept of Acids and Bases

Lewis proposed that acids are electron pair acceptors and bases are electron pair donors. For example, aluminium chloride, reacts with ammonia in the same way that a proton donor does. Using curved arrows to show the donation of the electron pair of ammonia (the Lewis base), it can be given as :



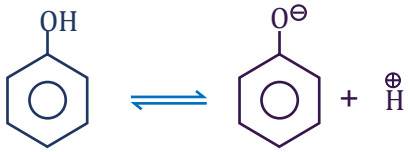
(I) Relative acidic strength of hydrocarbons :

Being most electronegative the sp hybridised carbon atom of ethyne or terminal alkyne polarizes its C–H bond to the greatest extent causing its H to be most positive therefore ethyne is most acidic. Hence acidic strength order is

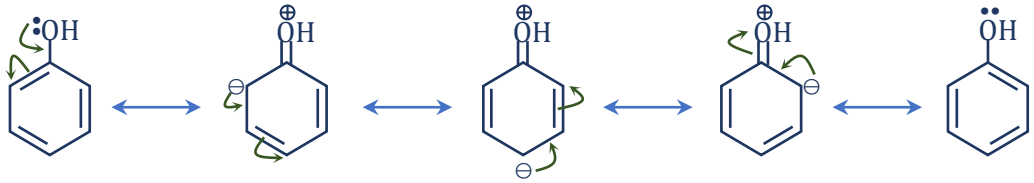
$HC \equiv CH > H_2C = CH_2 > H_3C - CH_3$

(II) Acidic Strength of Phenols :

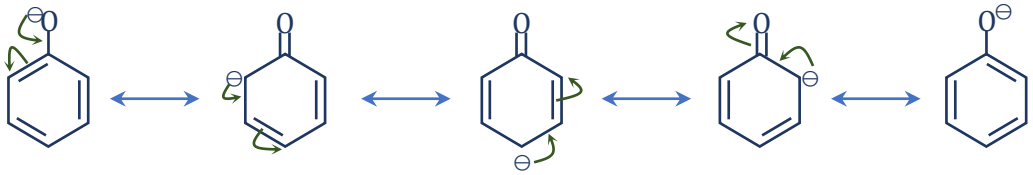
When phenol ionises, the phenoxide ion is more stabilised by resonance than the unionised phenol.



Resonating structures of phenol are :



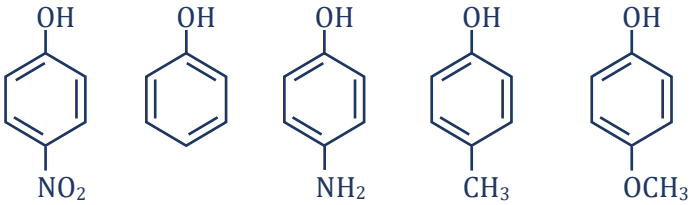
Resonating structures of phenoxide ion are :



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Groups which are – I, – M increases acidic character of phenol by effectively dispersing negative charge of phenoxide ion. However, +I and + M groups decreases acidic strength.

Example :

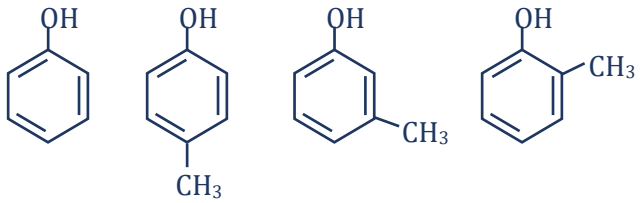


(I) (II) (III) (IV) (V)

Acid strength order : I > II > IV > V > III (–NO_2 show –M & –I while –OCH_3, –NH_2 show –I and +M)

So nitrophenol will be most acidic followed by phenol. Amongst cresol and methoxyphenol, methoxyphenol has + M effect of – OCH_3 which increases electron density hence decrease acidic strength.

Example:



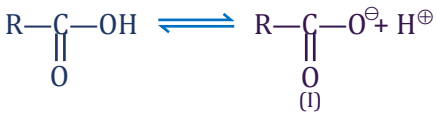
(I) (II) (III) (IV)

Acid strength order : I > III > II > IV

CH_3 have + I effect so all methyl phenols (cresols) are less acidic than phenol (I). Amongst cresols, para and ortho CH_3 are increasing the electron density due to their hyper conjugation but ortho isomer has viable + I effect also, which will help in destabilising phenoxide ion therefore o- is least acidic. Since at meta position only + I work it has least electron density amongst the cresols

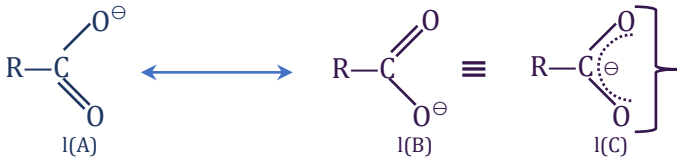
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(III) Acidic strength of Carboxylic Acids:

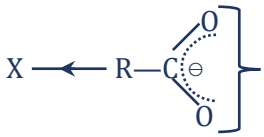


(I)

$RCOO^{\ominus}$ (I) exists as two equivalent canonical structures I(A) and I(B). This ion is resonance stabilised and resonance hybrid structure is I(C), while RCOOH has two non-equivalent resonating structures making it less stable.



Electron withdrawing group (–I effect) stablises the anion and hence, increases acidic nature.

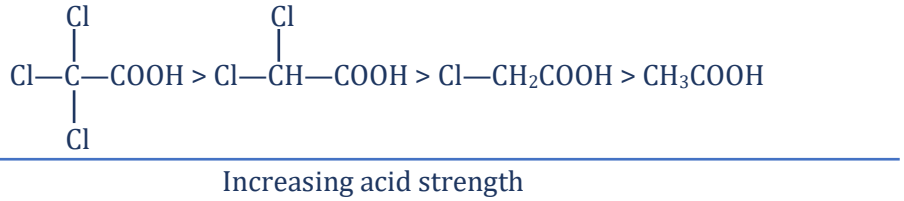


Ex. 1 $F—CH_2—COOH > Cl—CH_2COOH > Br—CH_2COOH > I—CH_2COOH$

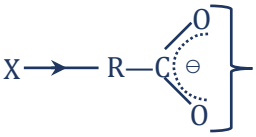
Increasing acid strength

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Ex. 2



Electron releasing group (+ I effect) destabilises the anion and hence decreases acidic nature.

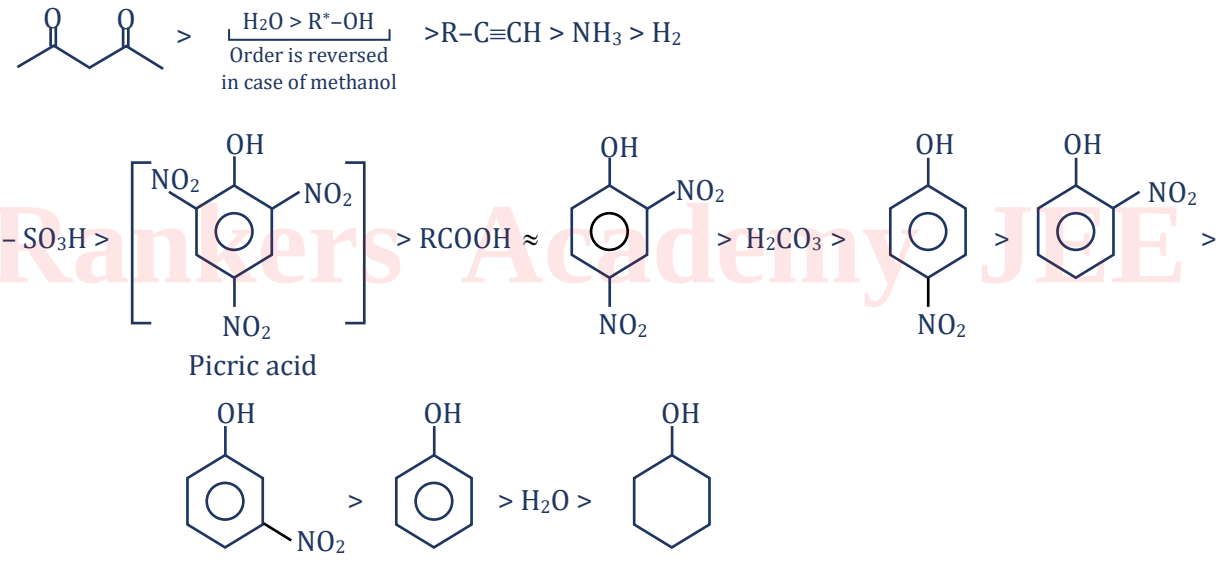


Ex. 3 $HCOOH > CH_3COOH > CH_3–CH_2–COOH$

Ex. 4



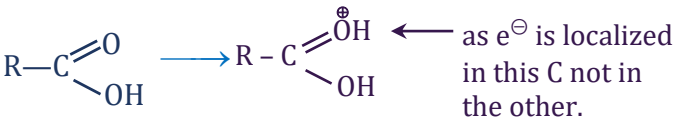
Important acid strength order:



Ex.1 Write conjugate acid of following :

(i) $H_2O \longrightarrow H_3O^{\oplus}$

(ii) $OH^{\ominus} \longrightarrow H_2O$

(iii)  ← as $e^{\ominus}$ is localized in this C not in the other.

(iv) $S^{2\ominus} \longrightarrow HS^{\ominus}$

(v) $HPO_4^{2\ominus} \rightarrow H_2PO_4^{\ominus}$

(vi) $NH_2–NH_2 \longrightarrow NH_2 – NH_3^{\ominus}$

(vii) 

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Ex.2 Write conjugate base of following:

(i) $H_2O \longrightarrow OH^{\ominus}$

(ii) $NH_3 \longrightarrow NH_2^{\ominus}$

(iii) 

(iv) 

(v) 

(vi) 

(vii) $H_2S \longrightarrow HS^{\ominus}$

Ex.3 A container with acid HA_1, HA_2, and HA_3 with pk_a : x, y, z respectively if y > z > x then.
$[HA_1 > HA_3 > HA_2]$

(a) $HA_1 + HA_2 \longrightarrow H_2A_1^{\oplus} + A_2^{\ominus}$

(b) $HA_1 + HA_3 \longrightarrow H_2A_3^{\oplus} + A_1^{\ominus}$

(c) $HA_2 + HA_3 \longrightarrow H_2A_2^{\oplus} + A_3^{\ominus}$

(d) $HA_2 + HA_3 \longrightarrow H_2A_3^{\oplus} + A_2^{\ominus}$

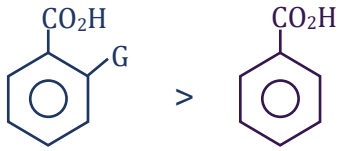
Option b & c are correct.

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SIR effect; Ortho effect and Mixed questions:

Ortho effect :

Ortho derivative of benzoic acid is generally more acidic than benzoic acid itself. It is called ortho effect.

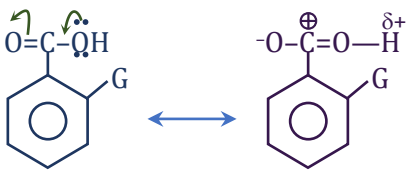


Explanation :

At ortho position when group G is attached, there will be steric repulsion between group G and $-CO_2H$ group, due to which $-CO_2H$ group is partially out of plane.

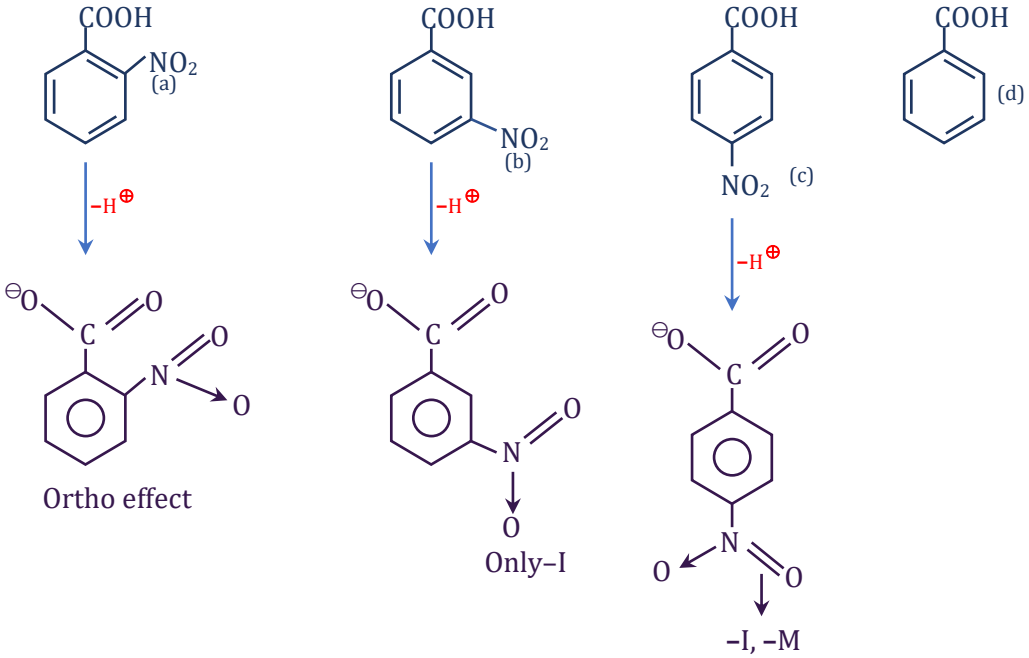
So, resonance of phenyl group w.r.t.  decreases and resonance of –OH group w.r.t.  increases,

+ve charge on oxygen atom increases, –I effect increases, acidity increases.



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Example : If substitutent is –I and –M ($-NO_2$)



K_a order = ortho > para > meta > benzoic acid

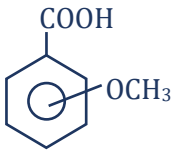
Example : If substitutent is (–I > +M) – Cl, Br, F, I



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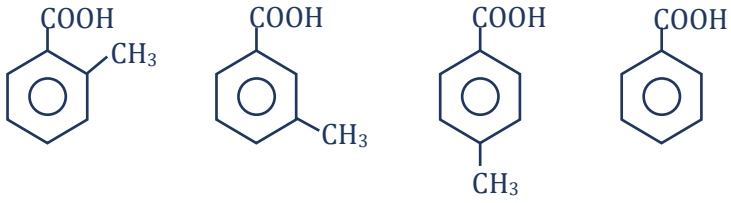
K_a order = ortho > meta > para > benzoic acid
acidity = a > c > b > d

Example : If substitutent is (+M > –I) $-OCH_3$



K_a order = ortho > meta > benzoic acid > para
Acidity = ortho > meta > benzoic acid > para

Example : If substitutent is (+I, +H) – R (Alkyl group)



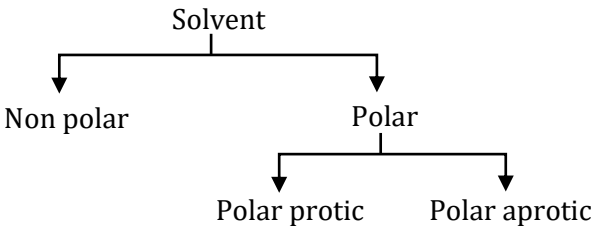
K_a order = ortho > benzoic acid > meta > para

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Electrophile, Nucleophile, solvent, leaving group, Nucleophilicity vs Basicity:

Solvent :

Solvent can be classified as:



1. Polar solvent :

Polar solvent has some overall dipole moment. They are of two types.

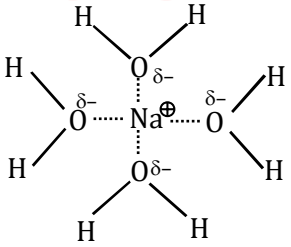
(i) Polar protic solvent : Solvents in which H atom is directly attached with highly electronegative atom. Polar protic solvents are capable of forming intermolecular hydrogen bonding.

In polar protic solvents cation are solvated by ion dipole interaction while an anion is solvated by hydrogen bonding. Polar protic solvent can give $H^{\oplus}$

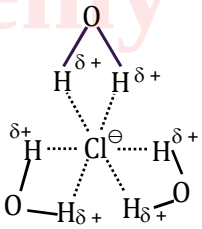
Example : H_2O, C_2H_5OH, CH_3COOH etc.

Polar protic solvent can solvate both cation as well as anion. Cations are solvated by ion dipole interaction with H_2O while anions are solvated by hydrogen bonding with H_2O. Solvation of NaCl in H_2O can be explained as

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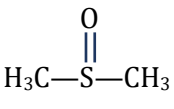
$Na^{\oplus}$ is solvated by ion-dipole interaction with HO_2



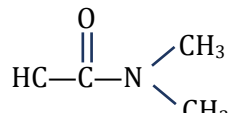
$Cl^{\ominus}$ is solvated by H-bonding with HO_2

(ii) Polar aprotic Solvents : These are the solvents in which H is not attached with highly electro negative atom. These solvents cannot give H^+

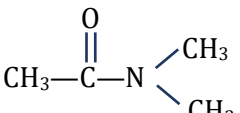
Examples :



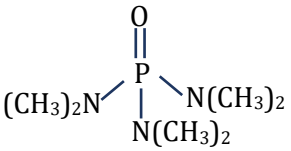
DMSO
(Dimethyl sulphoxide)



DMF
(Dimethyl formamide)



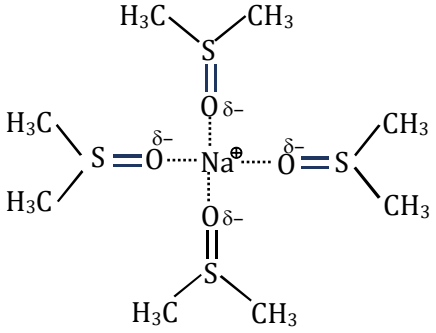
DMA
(Dimethyl acetamide)



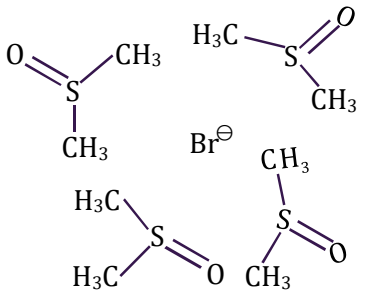
HMPA
(Hexamethylphosphoric triamide)

Polar aprotic solvent can solvate cation very well, while anion is not solvated. When a salt is added then cation is trapped by ion dipole interaction, while anion is not solvated

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$Na^{\oplus}$ is solvated by ion dipole interaction with DMSO



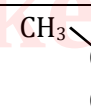
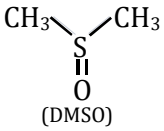
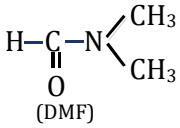
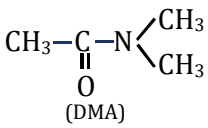
$Br^{\ominus}$ surrounded by solvent but not well solvated by DMSO

2. Non-Polar Solvents:

These solvent has overall zero dipole moment.

Examples : CCl_4, Benzene etc.

Some Common examples of solvents :

S.No.	Solvents	Polar protic	Polar aprotic
1.	H_2O	✓	–
2.	CH_3CH_2OH	✓	–
3.	CH_3CH_2OH	✓	–
4.	$CH_3–COOH$	✓	–
5.	NH_3	✓	–
6.	 (acetone)	×	✓
7.	 (DMSO)	×	✓
8.	 (DMF)	×	✓
9.	 (DMA)	×	✓

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Illustration 8

Which of the following is a non polar solvent ?

(A) H_2O (B)  (C) CCl_4 (D) Ethanol

Ans. (C)

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Solution:

H_2O,  & ethanol are polar solvent.

But, in CCl_4, net dipole moment is zero, so non-polar.

Attacking Reagents:

The species which attack on a substrate molecule or intermediate and form a product is called as attacking reagent.

These are of two types : (1) Electrophiles (2) Nucleophiles

Electrophiles & nucleophiles :

1. Electrophile :

An electrophile is a positively charged species or neutral molecule with electron deficient center. Electrophiles can act as Lewis acids.

Electrophiles are of following type :

(i) Positively charged electrophiles :

$H^{\oplus}$, $Br^{\oplus}$, $Cl^{\oplus}$, $I^{\oplus}$, $NO_2^{\oplus}$, $R_3C^{\oplus}$, $NO^{\oplus}$ etc.

(ii) Neutral Electrophiles : In these electrophiles central atom has an incomplete octet and at least one vacant orbital

SO_3, BF_3, $AlCl_3$, $ZnCl_2$, $FeCl_3$, RCOCl, $(RCO)_2O$

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2. Nucleophiles :

It is an electron rich species which has at least one lone pair of electrons. Nucleophile can be neutral or negatively charged. Nucleophile is always a Lewis base.

Example : $CN^{\ominus}$, $OH^{\ominus}$, $Br^{\ominus}$, $I^{\ominus}$, NH_3 , H_2O

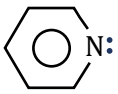
Nucleophile are of following types :

(i) Negatively charged nucleophiles :

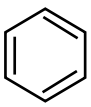
Example : $\overset{\ominus}{H}$, $\overset{\ominus}{O}H$, $\overset{\ominus}{O}—R$, $\overset{\ominus}{C}H_3$, $\overset{\ominus}{X}$, $\overset{\ominus}{S}H$, $R—\overset{\ominus}{S}$, $\overset{\ominus}{N}O_2$, $R—\overset{O}{\overset{\|}{C}}—\overset{\ominus}{O}$, etc

(ii) Neutral nucleophiles : In these nucleophiles central atom has complete octet and at least one lone pair of electron

Example : $R—\ddot{N}H_2$, $\ddot{N}H_3$, $\ddot{N}H_2—\ddot{N}H_2$, $H—\ddot{O}—H$, $R—\ddot{O}—H$, $R—\ddot{O}—R$,

$H—\ddot{S}—H$, $R—\ddot{S}—H$, $R—\ddot{S}—R$, $\ddot{P}H_3$, 

Organic compounds containing carbon-carbon double bonds can also behave as nucleophile.

$CH_2=CH_2$,  , $CH_2=CH—CH=CH_2$, etc.

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(iii) Ambident Nucleophile : Species having two nucleophilic sides, but only one can donate lone pair of electron at a time, known as ambident nucleophile.

$C^{\ominus}\equiv\ddot{N}$, $^{\ominus}O-\ddot{N}=O$, etc.

(iv) Amphiphile : Molecule containing a multiple bond between carbon and a more electronegative atom can act both as electrophiles or nucleophiles

$>C=O$ $CH_3-C\equiv N$

S.No.	Electrophile	Nucleophile
1	Accepts the electron pair	Supplies the electron pair
2	Electron deficient	Electron rich
3	Lewis acid	Lewis base
4	Possess an empty orbital to receive the electron pair	Possess an electron pair which is loosely held and can be supplied easily
5	Usually positively charged species	Usually negatively charged species
6	Forms an extra bond with the nucleophile	Increases its covalency by one unit

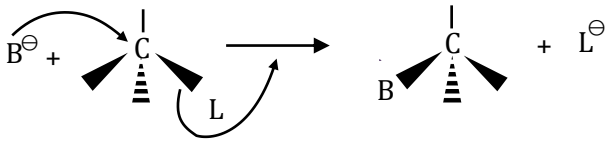
Nucleophilicity & basic strength:

Nucleophilicity is defined as the tendency of any species to give electron pair to an electron deficient center, while base strength is the ability of the species to remove $H^{\oplus}$ ion, from an acid

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Example : Consider a species $B^{\ominus}$

(i) It functions as a nucleophile is given as



(L-Leaving group)

(ii) It functions as base is indicated as

$B^{\ominus} + H-A \rightleftharpoons B-H + A^{\ominus}$

The nucleophilicity is determined by the kinetics of the reaction, which is reflected by its rate constant (k) while base strength is determined by the equilibrium constant, which is reflected by its K_b.

Criteria for Nucleophilicity: The factors which increases electron density at donor atom increases nucleophilicity.

The more polarisable donor atom is a better nucleophile. Therefore, large size of donor atom increases nucleophilicity.

Effect of the solvent : In polar protic solvent large nucleophiles are good, and the nucleophilicity of halide ions follows the order as :

$F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$

In DMSO, the relative order of nucleophilicity of halide ions is given as

$F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

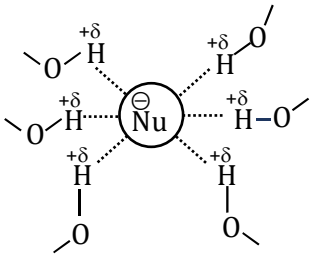
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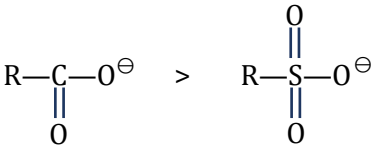
Also, base strength of halides follows the same order in DMSO but in polar protic solvent like H_2O they follow the reverse order of nucleophilicity.

This effect is related to the strength of the interaction between nucleophile and solvent molecules of polar protic solvent forms hydrogen bond to nucleophiles in the following manner :

Relative nucleophilicity in polar protic solvent

$$\overset{\ominus}{S}H > \overset{\ominus}{I} > \overset{\ominus}{O}H > \overset{\ominus}{N}_3 > \overset{\ominus}{Br} > \overset{\ominus}{Cl} > \overset{\ominus}{F} > H_2O$$



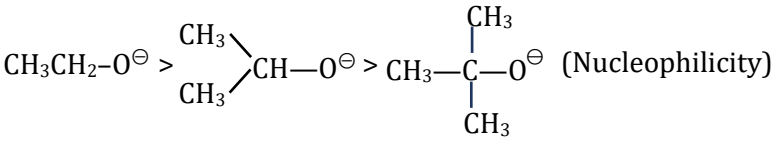


$$HO^{\ominus} > H_2O$$

$$NH_2^{\ominus} > NH_3$$

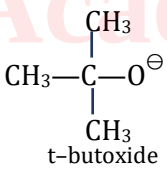
$$CF_3SO_3^{\ominus} < PhCOO^{\ominus} < PhO^{\ominus} < RO^{\ominus}$$

Steric effects on nucleophilicity :

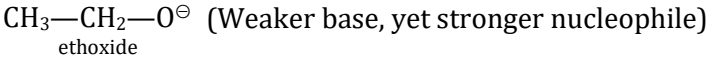


Due to steric crowding nucleophilicity decreases.

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(Stronger base, yet weaker nucleophile cannot approach the carbon atom so easily)



Periodicity : Nucleophilicity decreases from left to right in a period and base strength also decreases.

$$CH_3^{\ominus} > NH_2^{\ominus} > OH^{\ominus} > F^{\ominus}$$

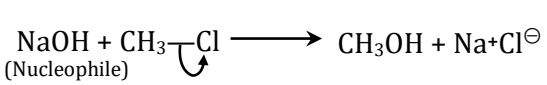
In a group, nucleophilicity increases from top to bottom due to increases in size of donor atom, but base strength decreases from top to bottom.

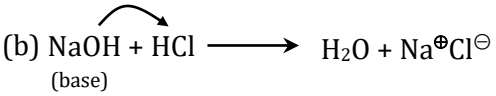
Acid strength : HI > HBr > HCl > HF

Base strength : $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

Nucleophilicity : $F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$

It is the tendency to donate electron pair to $H^{\oplus}$ ion.

(a) 

(b) 

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NUCLEOPHILICITY vs BASIC STRENGTH

S.NO.	Nucleophilicity	Basic strength	Remarks
1.	$CH_3^{\ominus} > NH_2^{\ominus} > OH^{\ominus} > F^{\ominus}$	$CH_3^{\ominus} > NH_2^{\ominus} > OH^{\ominus} > F^{\ominus}$	If donor atoms belong to same period, nucleophilicity and basic strength order are same
2.	$SiH_3^{\ominus} > PH_2^{\ominus} > SH^{\ominus} > Cl^{\ominus}$	$SiH_3^{\ominus} > PH_2^{\ominus} > SH^{\ominus} > Cl^{\ominus}$	If donor atoms belong to same period, nucleophilicity and basic strength order are same
3.	$F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$	$F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$	In a group nucleophilicity increases while basic strength decreases. on moving top to bottom.
4.	$OH^{\ominus} < SH^{\ominus}$	$OH^{\ominus} > SH^{\ominus}$	In a group nucleophilicity increases while basic strength decreases. on moving top to bottom.
5.	$RO^{\ominus} < RS^{\ominus}$	$RO^{\ominus} > RS^{\ominus}$	In a group nucleophilicity increases while basic strength decreases. on moving top to bottom.

NOTE :

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1. Nucleophilicity depends on the nature of solvent.
2. Anions are better nucleophile than their neutral molecule.
3. Nucleophilicity increases if adjacent atom of nucleophilic side has lone pair, because lone pairs repels each other. Example $\ddot{N}H_2 - \ddot{N}H_2$.

Leaving Group Ability/Nucleofugacity :

Good leaving groups are those that become the most stable ion after they leave, because leaving group generally leave as a negative ion, so those leaving group are good, which stabilise negative charge most effectively and weak base do this best, so weaker is the base better is the leaving group.

– The leaving group should have lower bond energy with carbon.

– Negative charge should be more stable either by dispersal or delocalization.

Strongly basic ions rarely act as leaving group →

$\overset{\ominus}{Br} + R-OH \xrightarrow{X} R-X + \overset{\ominus}{O}H$ (strong base / poor leaving group)

$\overset{\ominus}{N}u + CH_3-CH_3 \xrightarrow{X} CH_3-Nu + \overset{\ominus}{C}H_3$ (It is not a leaving group)

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Order of leaving ability of some groups:

1. $R-\overset{\oplus}{N}\equiv N > R-\overset{\oplus}{O}(H)-R' > R-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-C_4F_9 > R-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-CF_3$

2. $R-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-O-C_6H_4-Br > R-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-O-C_6H_4-CH_3$

3. $R-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-C_6H_5 > R-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-CH_3 > RI > R-Br > RCl > RF$

4. $R-O-\overset{\overset{O}{\|}}{C}-CH_3 > R-O-H > R-O-R > R-H$

where group attached with R is a leaving group

Illustration 9

Which of the following species is an electrophilic in nature :-

(A) BH_3 (B) $H_3O^{\oplus}$ (C) $^{\oplus}NO_2$ (D) $Cl^{\oplus}$

Ans. (A, C, D)

Solution:

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(A) BH_3 due to empty p orbital, BH_3 is electrophilic in nature

(B) $H_3O^{\oplus}$ even through contains positive charge but not electrophilic in nature due to unavailable free orbital.

(C) $\overset{\oplus}{N}O_2, O\leftarrow\overset{\oplus}{N}=O \xrightarrow{Nu^{\ominus}} O\leftarrow\underset{\underset{Nu}{|}}{N}-O^{\ominus}$

So $\overset{\oplus}{N}O_2$ is electrophilic in nature

(D) $Cl^{\oplus}$ due to empty p orbital electrophilic in nature.

Basic strength + SIP Effect:

(I) Aliphatic Amines

Ease of protonation is the basic nature. Increasing strength of nitrogenous bases is related to readiness with which they are prepared to take up protons.

Consider the following molecules

$\ddot{N}H_3$, $Me \rightarrow NH_2$, $Me \rightarrow \underset{\underset{Me,}{\uparrow}}{NH}$ $Me \rightarrow \underset{\underset{Me}{\uparrow}}{N} \leftarrow Me$

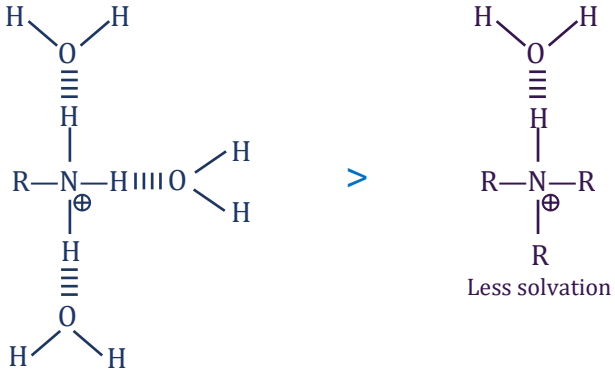
By considering + I effect of methyl in above example we may expect basic nature as

$NH_3 < MeNH_2 < Me_2NH < Me_3N$ **(Which is not true always)**

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This is due to the fact that basic strength of an amine in water is determined not only by ease of electron donation (protonation) of N atom but also by the extent to which cation so formed can undergo **SOLVATION** and become stabilised by H atom attached to N atom.

Greater is possibility of solvation via H bonding by water more will be its basicity but Alkyl groups are hydrophobic and inhibits H bonding hence solvation decreases and basicity decreases.



Thus, due to two opposite effects i.e. solvation of cation and +I effect, the jumbled order comes to be

$Me_2NH > Me - NH_2 > Me_3N > NH_3$

It is also interesting to note that in gas phase basicity order of amines follows the usual order of $Me_3N > Me_2NH > MeNH_2 > NH_3$ (Since only + I effect works and no solvation effect persists).

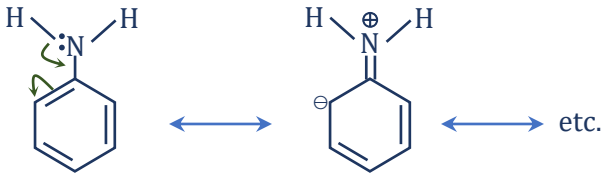
Similarly, on the same reasons ethyl amines and other amine follows the following order for basic strength:

$Et_3N > Et_2NH > EtNH_2 > NH_3$ (gas phase)

$Et_2NH > Et_3N > EtNH_2 > NH_3$ (water as solvent)

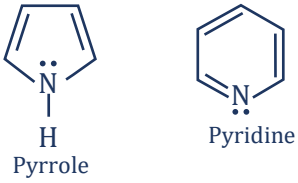
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(II) Aromatic amines ($Ph-\ddot{N}H_2$) or Anilines :

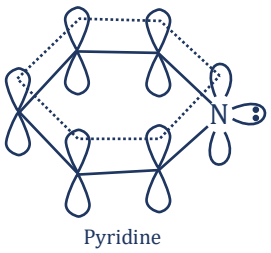


When the lone pair lies in conjugation with a multiple bond, system can get resonance stabilisation. Aniline is a weaker base than NH_3 because it has delocalised lone pair.

Example :



In pyrrole lone pairs are involved in resonance therefore it is less basic. But in pyridine lone pairs are in perpendicular plane of π orbitals therefore not involved in resonance



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Basic Strength :

* Acid amides are weaker bases
* Ability to donate $e^{\ominus}$
* Check stability of conjugate acid

1.

I	II	III	IV
$I^{\ominus}$	$Cl^{\ominus}$	$Br^{\ominus}$	$F^{\ominus}$
↓	↓	↓	↓
HI	HCI	HBI	HF

IV > III > II > I as the acidic character decreases from I→IV, the basic character ↑

2.

I	II	III	IV
$^{\ominus}CH_3$	$^{\ominus}NH_2$	$^{\ominus}OH$	$^{\ominus}F$
↓	↓	↓	↓
CH_4	NH_3	H_2O	HF

I > II > III > IV → as E.N. ↑ , –ve charge becomes more stable and basic ↓

3. I II

$RO^{\ominus}$ > $RS^{\ominus}$

↓ ↓

ROH RSH

I > II as smaller the size, more will be the charge density.

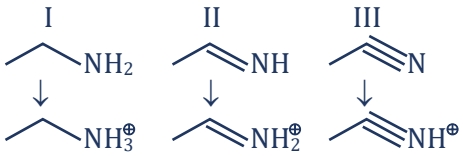
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4.

I	II
R_3N	R_3P
↓	↓
$^{\oplus}NR_3H$	$^{\oplus}PR_3H$

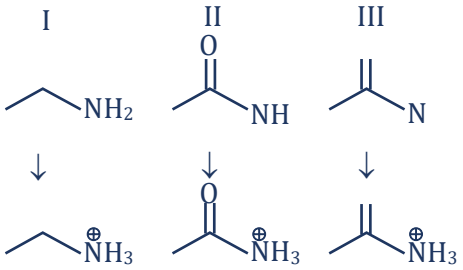
I > II, smaller the size, better the overlapping

5.



I > II > III as s character ↑, acidity ↑ Basicity ↓

6.



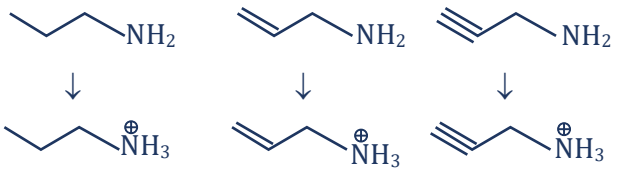
I > III > II, I - Localized

II - Dispersed on O ⇒ highly delocalize lone pair

III - Dispersed on C ⇒ moderately localize.

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7. (I) (II) (III)



> II > III as +I effect decreases from sp^3 C → sp C.

8. (I) (II) (III)

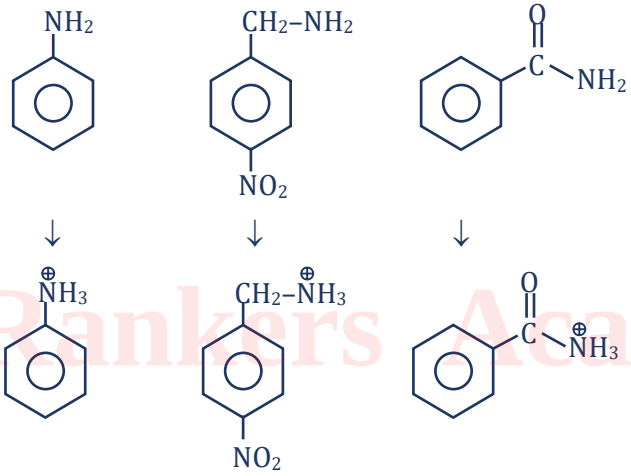
$HC \equiv C^{\ominus}$ $CH_2=CH^{\ominus}$ $CH_3-CH_2^{\ominus}$

↓ ↓ ↓

$HC \equiv CH$ $CH_2 = CH_2$ $CH_3- CH_3$

III > II > I as s corrector decreases from I → III.

9. (I) (II) (III)



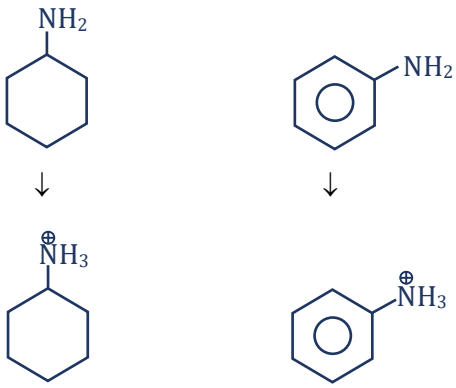
II > I > III

II–Localized L.P.

I–$e^{\ominus}$ dispersed on C

III–$e^{\ominus}$ dispersed on O

10. (I) (II)



I > II due to localized L.P.

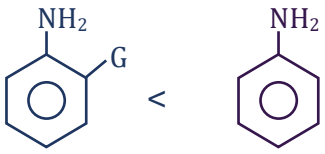
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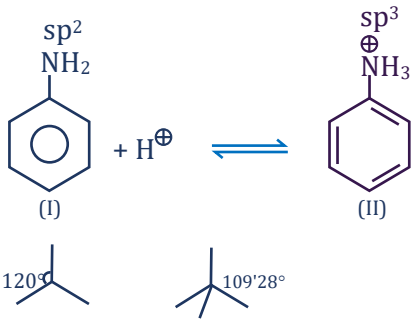
SIP Effect :

Ortho effect :

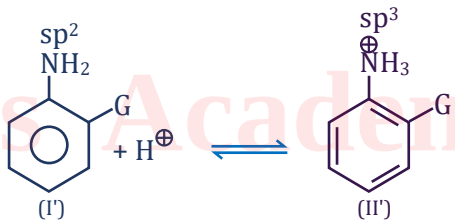
Ortho derivative of aniline is generally less basic than aniline itself. It is called ortho effect.



during protonation N change its hybridisation from sp^2 to sp^3. Steric crowding around N increases. If G is occupied at ortho position then steric crowding further increases, stability of protonated product decreases, basicity decreases.



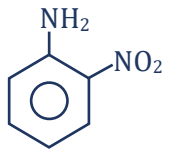
Change of hybridization causes steric repulsion.

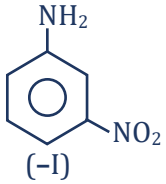


steric crowding further increases, stability of II' further decreases basicity order to I' < I

Example : Arrange the following in their basic strength

(i) G = (–M, –I); NO_2

(A)  (–I, –M ortho effect)

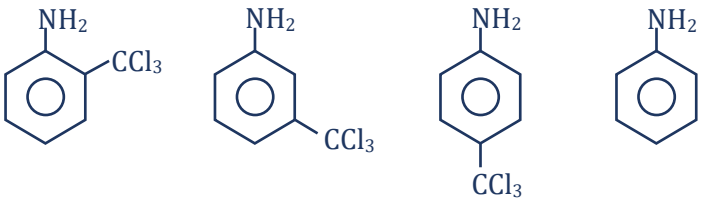
(B)  (–I)

(C)  (–I,–M)

(D) 

Solution : (Aniline > m > p > o).

(ii) G = (–I); CCl_3



Solution : (Aniline > m > p > o).

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Basic idea of carbene and nitrene:

Carbene:

A carbene is a molecule containing a neutral carbon atom with a valency of two and two unshared valence electrons. The general formula is R–:C–R'



Nitrene:

A nitrene (R–:Ṅ·) is the nitrogen analogue of a carbene. The nitrogen atom is uncharged and monovalent, so it has only 6 electrons in its valence level—two covalent bonded and four non-bonded electrons. It is therefore considered an electrophile due to the unsatisfied octet. A nitrene is a reactive intermediate and is involved in many chemical reactions.

Acid Base Equilibrium; $NaHCO_3$ test and Reaction of Na, aq. NaOH, 5% HCL:

Ex 1 : Feasible reaction is/are :

(a) $R–C\equiv CH + NaOH \longrightarrow RC\equiv C–Na^{\oplus} + H_2O$

(b) $RC\equiv CH + KOH \longrightarrow R–C\equiv C–K^{\oplus} + H_2O$

(c) $RC\equiv CH + NaNH_2 \longrightarrow R–C\equiv C–Na^{\oplus} + NH_3$

(d) $RC\equiv CH + NaH \longrightarrow R–C\equiv C–Na^{\oplus} + \frac{1}{2}H_2$

(e) $RC\equiv CH + NaHCO_3 \longrightarrow R–C\equiv C–Na^{\oplus} + H_2CO_3$

Compare these. clearly c, d are feasible

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Ex.2: Product when R–COOH reacts with aq. $NaHCO_3$:

$$R–COOH + NaHCO_3 \longrightarrow R–\underset{\underset{O}{\|}}{C}–ONa + [H_2O + CO_2]$$

Ex.3: Gases evolved when $NaH^{14}CO_3$ is added in a container having R–COOH and $R–SO_3H$:

(i) $R–COOH + NaH^{14}CO_3 \longrightarrow RCOO–Na^{\oplus} + H_2O + {}^{14}CO_2$

(ii) $R–SO_3H + NaH^{14}CO_3 \longrightarrow R–SO_3HNa + H_2O + {}^{14}CO_2$

Ex.4: Reaction between p-nitrophenol and aq. $NaHCO_3$:



Ex.5 Find no. of compounds reacting with:

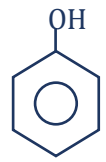
(i) aq. NaOH

(ii) aq. $NaHCO_3$

(iii) both in NaOH and $NaHCO_3$

(iv) give effervescence of CO_2 with $NaHCO_3$.

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(a) 

(NaOH)

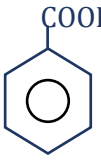
(b) 

($NaOH + NaHCO_3$)

(c) 

($NaOH + NaHCO_3$)

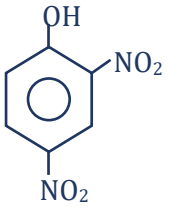
(d) 

(e) 

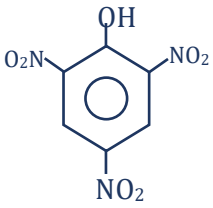
($NaOH + NaHCO_3$)

(f) 

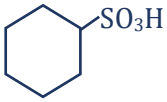
($NaOH + NaHCO_3$)

(g) 

($NaOH + NaHCO_3$)

(h) 

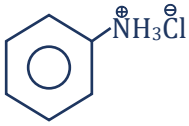
($NaOH + NaHCO_3$)

(i) 

($NaOH + NaHCO_3$)

(j) HCl
($NaOH + NaHCO_3$)

(k) H_2SO_4c
($NaOH + NaHCO_3$)

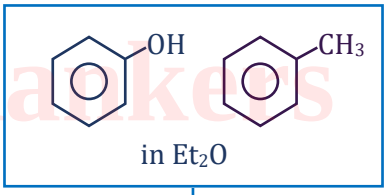
(l) 

($NaOH + NaHCO_3$)

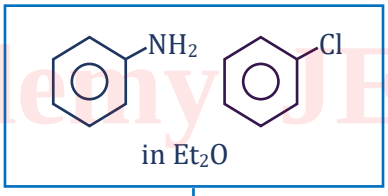
(m) 

(n) 

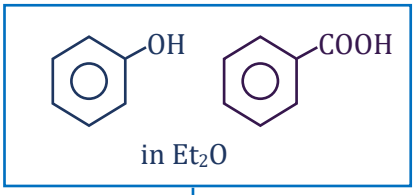
Ex.6 How can we separate following binary mixture by acid-base extraction method.

(i) 

aq. $NaNH_2$
Ph – ONa
Ph – CH_3
aq. H_2SO_4
Et_2O
Ph – OH
Ph – CH_3

(ii) 

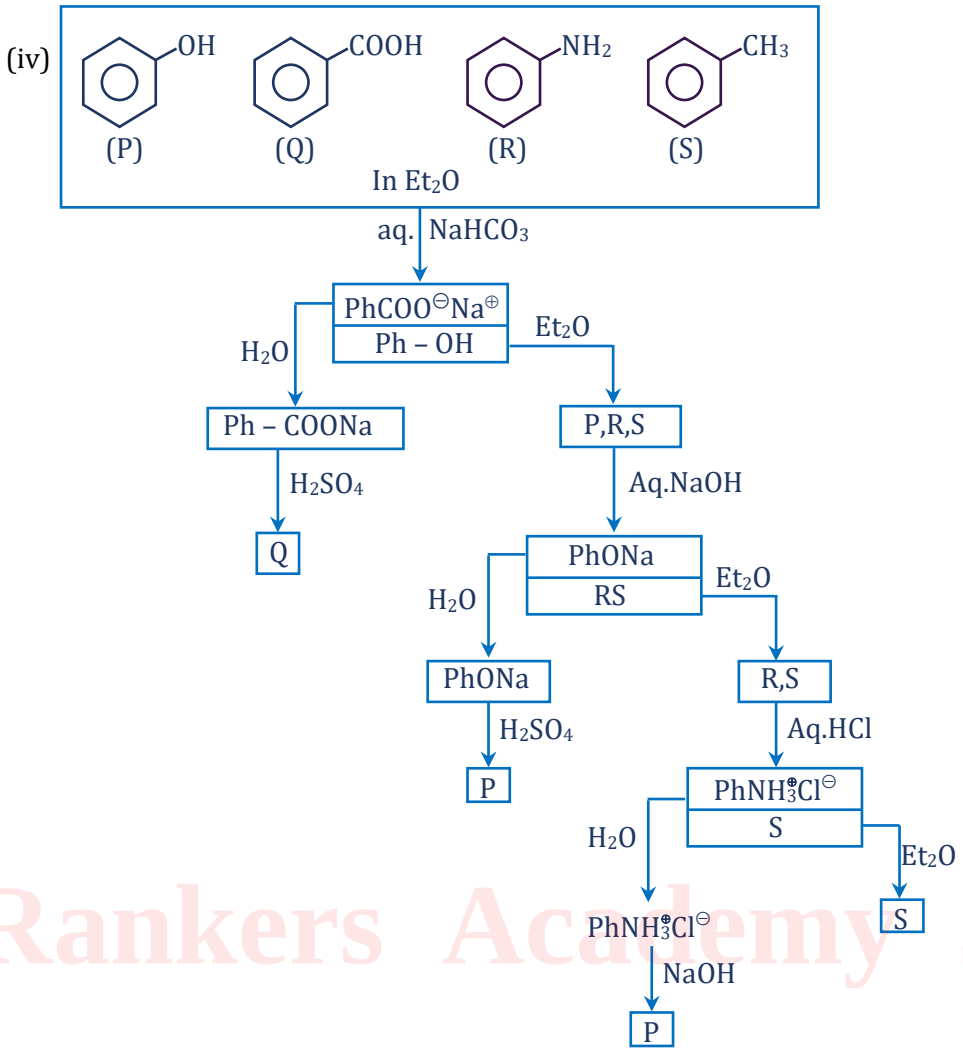
aq. HCl
Ph – $NH_3^{\oplus}Cl^{\ominus}$
Ph – Cl
aq. NaOH
Et_2O
Ph – NH_2
Ph – Cl

(iii) 

aq. $NaHCO_3$
Ph – COONa
Ph – OH
aq. HCl
Ph – COOH
Ph – OH

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Purification of Organic Compounds, Sublimation, Crystallization, Distillation, Steam Distillation, Differential Extraction:

The common techniques used for purification are as follows :

(i) Sublimation :

Change from solid to vapour state without passing through liquid state is known as sublimation.

Sublimation is used to separate sublimable compounds from nonsublimable impurities.

(ii) Crystallisation:

Principle : Based on the difference in solubility of organic compound and impurity in a suitable solvent.

- The impure compound is dissolved in a solvent in which it is sparingly soluble at room temperature but appreciably soluble at higher temperature.
- The solution is concentrated to get a nearly saturated solution.
- On cooling the solution, the pure compound crystallised out and is removed by filtration.
- Example : Sugar having impurity of common salt can be crystalised in hot ethanol.

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(iii) Distillation:

This important method is used to separate (i) volatile liquids from nonvolatile impurities and (ii) the liquids having sufficient difference in their boiling points. Liquids having different boiling points vaporise at different temperatures. The vapours are cooled, and the liquids so formed are collected separately.

Following distillation techniques are used for purification of organic compound :

(a) Simple distillation :

- Use for separating compounds which have boiling point difference of atleast 20° C.
- Chloroform (b.p 334 K) and aniline (b.p. 457 K) are easily separated by the technique of simple distillation.

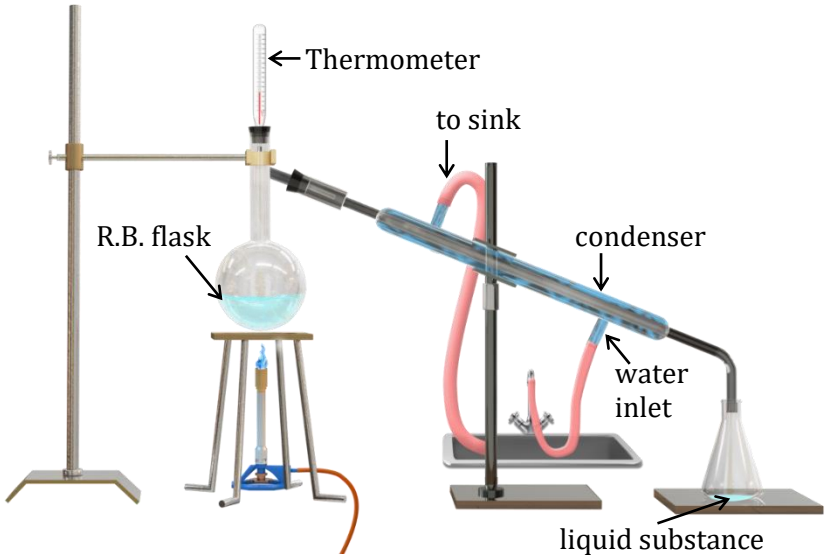


Figure : Simple distillation.

The vapours of a substance formed are condensed and the liquid is collected in conical flask.

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(b) Fractional distillation:

- It is used when two liquids do not have much difference in boiling point.

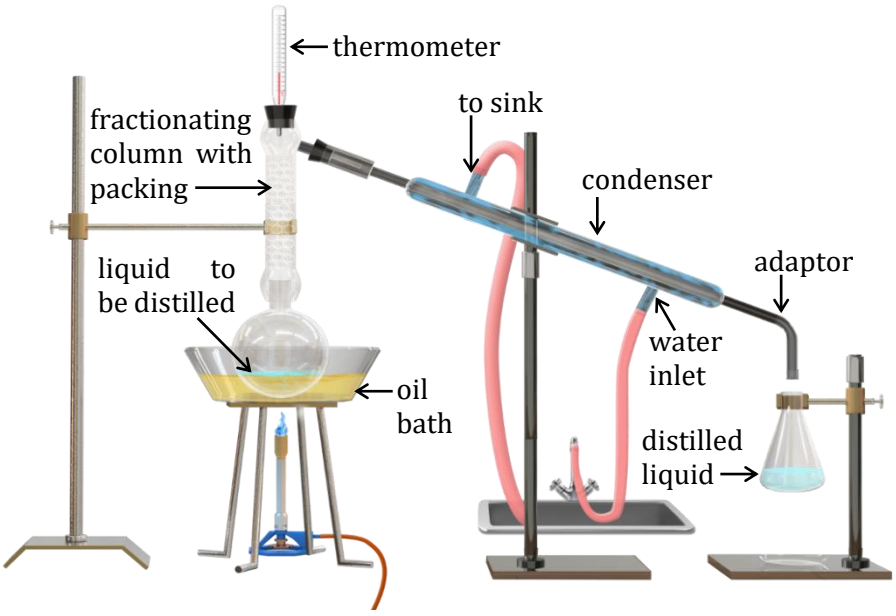


Figure : Fractional distillation.

The vapours of lower boiling fraction reach the top of the column first followed by vapours of higher boiling fractions.

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- In this process a fractionating column is fitted over the mouth of round bottom flask through which vapours will passed.
- In the fractionating column, the more volatile compounds travel faster and reach the top of the column first. Less volatile compounds condense first in the fractionating column.
- Fractionating columns provide many surfaces for heat exchange between the ascending vapour and the descending condensed liquid.
- The vapor thus become richer in low boiling component and passed through the condenser and the pure liquid is collected in a receiver.
- The remaining liquid in the distillation flask gets enriched in high boiling component.
- Fractional distillation is also dependent on length of fractionating column.
- Example : Crude Oil refining

(c) Distillation under reduce pressure:

- It is used to purify liquid having high boiling points and compounds which decompose at or below their boiling point.
- Under the condition of reduced pressure, the liquid will boil at a temperature lower than its boiling point and as a result the liquid will not degrade.
- Example : Glycerol (Glycerol + Spent lye) in soap industry

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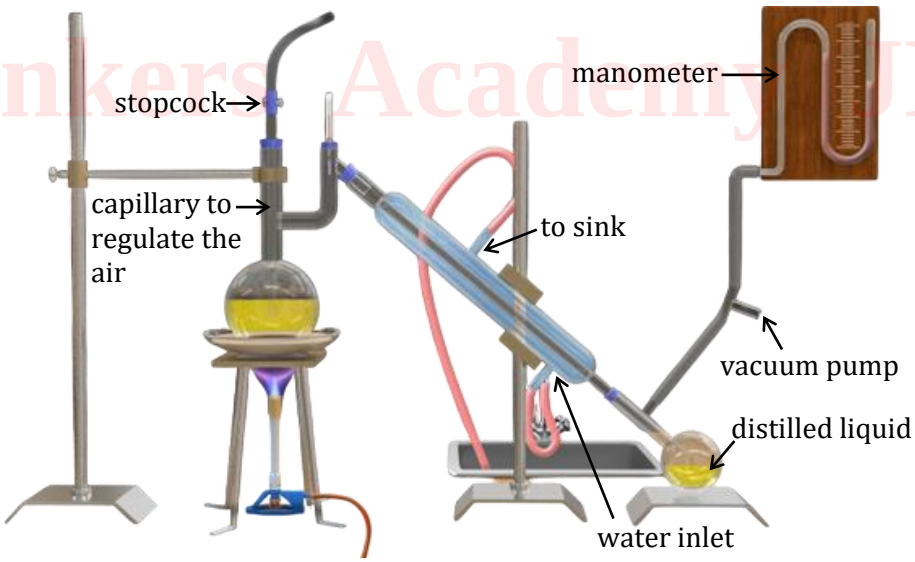


Figure : Distillation under reduced pressure.

A liquid boils at a temperature below its vapour pressure by reducing the pressure.

(d) Steam distillation : Used to purify compounds which are steam volatile and insoluble into H_2O.

- In steam distillation, steam from a steam generator is passed through a heated flask containing the liquid to be distilled.
- The compound is later separated from water using a separating funnel.
- **Example** : Aniline is separated from an aniline-H_2O mixture. Ortho and para nitrophenol.

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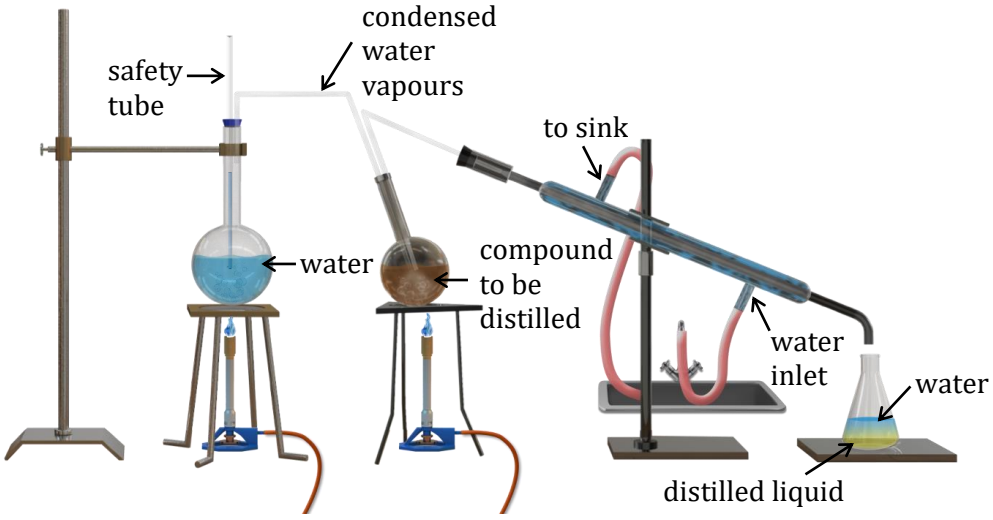


Figure : Steam distillation.
The vapours of lower boiling fraction reach the top of the column first followed by vapours of higher boiling fractions.

(iv) Differential extraction:

- The process of isolating an organic compound from its aqueous solution by shaking it with a suitable organic solvent is called differential extraction.
- A separatory funnel is used for separating them.
- Common organic solvents are ether, benzene, and acetone.
- Water and organic solvents should not be miscible.
- Organic compounds more soluble in organic solvents but less soluble in water. Now an aqueous solution of organic compound is mixed with organic solvent in separating funnel so organic compound dissolved in organic solvent and get separated from aqueous layer.
- Now organic compound in solvent layer get extracted and organic solvent later removed by evaporation to get back the compound.
- **Example** : Benzoic acid from its aqueous solution in benzene solvent.

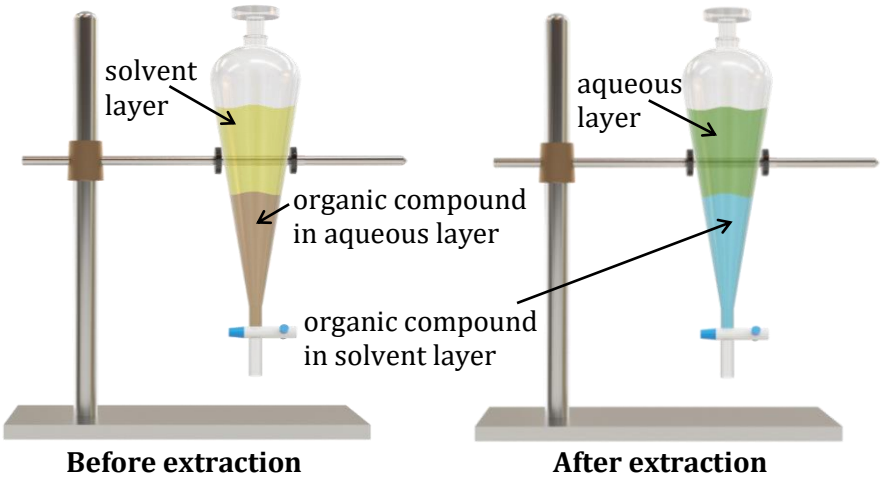


Figure : Differential extraction.
Extraction of compound takes place based on difference in solubility

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Chromatography

- It is used to separate the mixture into their components, purify compounds and test the purity of compounds.
- Stationary Phase- The mixture is applied onto a substance which may be liquid or solid is called stationary phase.
- Mobile Phase- It is a pure solvent or mixture of solvents that moves slowly over the stationary phase, which is a moving phase.
- Based on the principles involved chromatography is categorised as :
  1. **Adsorption Chromatography**
     (a) Column Chromatography
     (b) Thin Layer Chromatography
  2. **Partition Chromatography**
     Paper chromatography is a type of partition chromatography.

1. Adsorption chromatography:

- It is based on the fact that different compounds are adsorbed on an adsorbent to different degrees.
- It is of two types:

(a) Column chromatography:

- Stationary Phase- Silica Gel/Alumina
- Mobile Phase- Solvent or mixture of solvent which is allowed to move. (Top to Down)
- In column chromatography, a column of glass tube is packed with adsorbent (silica and glass) and the column is packed with glass wool and fitted with a stopcock at its lower end.
- The mixture is placed on the top of the adsorbent column and packed in a glass tube.
- Now, the solvent is allowed to flow down the column slowly. Depending upon the degree to which the compounds are adsorbed, complete separation takes place.
- The most readily adsorbed (High polarity) substances are retained near the top and others come down to various distances in the column.

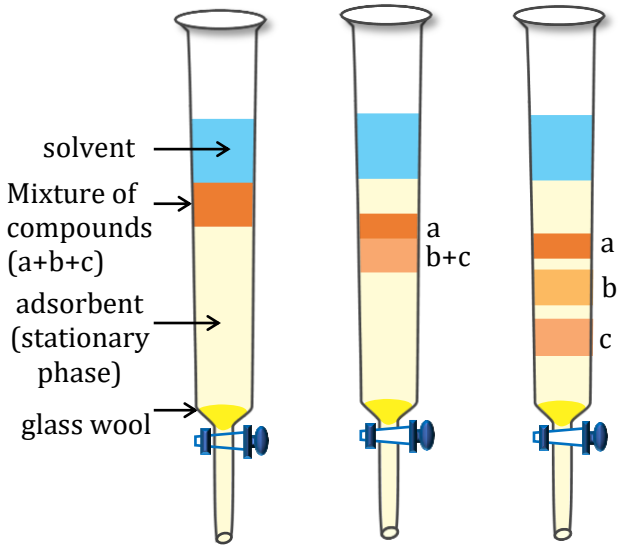


Figure : Column chromatography.

Different stages of separation of components of a mixture.

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(b) Thin layer chromatography:

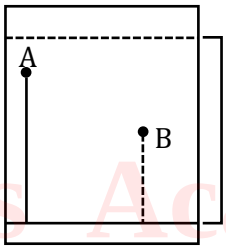
- Stationary Phase- Silica Gel/ Alumina
- Mobile Phase- Solvent or mixture of solvent

(Down to Top, Capillary action)

- It is also used to separate the mixture of organic compounds over a thin layer of an absorbent coated on a glass plate.
- Prepare a thin layer of adsorbent (alumina/silica) on a glass plate known as thin layer chromatography plate.
- Now, a baseline is drawn from 2 cm up from the bottom and a small spot of mixture is placed through a capillary tube on the TLC plate.
- Now, the TLC plate has been placed in a jar, containing the solvent. The solvent rises up (Capillary action) the plate, components of the mixture move up along with the solvent at different distances depending upon their degree of adsorption and separation takes place.
- The relative adsorption of each component of the mixture is expressed in terms of its retardation factor (R.f.) value.

$$\text{R.F.} = \frac{\text{Distance travelled by mixture from baseline}}{\text{Distance travelled by solvent from baseline}}$$

(Solvent Front)



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Here, Polarity A < B

NOTE :

- If spots are colourless, then they can be detected by putting the plate under ultraviolet light.
- Place the plate in a jar containing iodine so that compounds adsorb iodine will show up as brown spots.

2. Partition chromatography:

- Stationary Phase- Water trapped in chromatography paper.
- Mobile Phase- Solvent or mixture of solvent.
- Partition chromatography is based on continuous differential partitioning of components of a mixture between stationary and mobile phases.
- Paper chromatography is a type of partition chromatography.
- In paper chromatography, chromatography paper (a special quality paper) has been used who contains water trapped in it, which acts as a stationary phase.
- Now, a baseline is drawn from 1 cm up from the bottom and a small spot of the mixture is placed through a capillary tube on chromatography paper.
- Now, chromatography paper is suspended in a suitable solvent or a mixture of solvents.
- The paper selectively retains different components according to their differing partitions in the two phases.
- The spots of the separated coloured compounds are visible at different heights from the position of initial spot.

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Element Detection; Detection of Carbon and Hydrogen; Detection of Nitrogen, Sulphur, Halogens and Phosphorus; Lassaigne's Test; Beilstein Test:

Detection of carbon and hydrogen:

Carbon and hydrogen are detected by heating the compound with copper (II) oxide. Carbon present in the compound is oxidised to carbon dioxide (tested with limewater, which develops turbidity) and hydrogen to water (tested with anhydrous copper sulphate, which turns blue).

$C + 2CuO \xrightarrow{\Delta} 2Cu + CO_2$

$2H + CuO \xrightarrow{\Delta} Cu + H_2O$

$CO_2 + Ca(OH)_2 \xrightarrow{\Delta} CaCO_3 \downarrow + H_2O$

$5H_2O + \underset{\text{White}}{CuSO_4} \xrightarrow{\Delta} \underset{\text{Blue}}{CuSO_4.5H_2O}$

Detection of other elements:

Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by "Lassaigne's test". The elements present in the compound are converted from their covalent form to their ionic form by fusing the organic compound with sodium metal.

Following reactions take place: C, N, S and X come from organic compound.

Cyanide, sulphide and halide of sodium formed by sodium fusion are extracted from the fused mass by boiling it with distilled water. This extract is known as sodium fusion extract or Lassaigne solution.

$Na + C + N \xrightarrow{\Delta} NaCN$; $2Na + S \xrightarrow{\Delta} Na_2S$

$Na + X \xrightarrow{\Delta} NaX$; $Na + C + N + S \xrightarrow{\Delta} NaSCN$

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(i) Test for nitrogen

The sodium fusion extract is boiled with iron (II) sulphate and then acidified with dilute sulphuric acid. The formation of Prussian blue or green colour confirms the presence of nitrogen.

Alternatively, $FeCl_3$ and dil. HCl may be added.

$6NaCN + FeSO_4 \longrightarrow Na_4[Fe(CN)_6]$

$Na_4[Fe(CN)_6] + 4Fe^{3+} \xrightarrow{xH_2O} \underset{\text{Ferric ferrocyanide (Prussian blue)}}{Fe_4[Fe(CN)_6]_3.xH_2O}$

(ii) Test for sulphur

(a) The sodium fusion extract is acidified with acetic acid and lead acetate is added to it. A black precipitate of lead sulphide indicates the presence of sulphur.

$Na_2S + (CH_3COO)_2Pb \longrightarrow PbS$ (Black)

(b) On treating sodium fusion extract with sodium nitroprusside, appearance of a violet colour indicates the presence of sulphur.

$Na_2S + Na_2[Fe(CN)_5NO] \longrightarrow Na_4[Fe(CN)_5NOS]$

Sodium nitroprusside — Sodium thionitroprusside (Violet/Purple)

(c) In case, nitrogen, and sulphur both are present in an organic compound, then sodium thiocyanate is formed. It gives blood red colour with neutral $FeCl_3$.

$Na + C + N + S \longrightarrow NaSCN$; Neutral $FeCl_3 + NaSCN \longrightarrow Fe(SCN)_3$(Blood red)

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(iii) Test for halogens

The sodium fusion extract is acidified with nitric acid and then treated with silver nitrate. A white precipitate, soluble in ammonium hydroxide shows the presence of chlorine, a yellowish precipitate, sparingly soluble in ammonium hydroxide shows the presence of bromine and a yellow precipitate, insoluble in ammonium hydroxide shows the presence of iodine.

NOTE :

The sodium fusion extract is first boiled with concentrated nitric acid to decompose cyanide or sulphide of sodium formed during Lassaigne's test. These ions would otherwise interfere with silver nitrate test for halogens.

$NaX + AgNO_3 \longrightarrow AgX\downarrow$

(iv) Test for phosphorus

The compound is heated with an oxidising agent (sodium peroxide). The phosphorus present in the compound is oxidised to phosphate. The solution is boiled with nitric acid and then treated with ammonium molybdate. A yellow colouration or precipitate indicates the presence of phosphorus.

$Na_3PO_4 + 3HNO_3 \longrightarrow H_3PO_4 + 3NaNO_3$

$H_3PO_4 + 12(NH_4)_2MoO_4 + 21\ HNO_3 \longrightarrow (NH_4)_3PO_4.\ 12\ MoO_3 + 21\ NH_4NO_3 + 12H_2O$

Ammonium Molybdate — Ammonium Phosphomolybdate

(A) ELEMENTAL ANALYSIS Lassaigne method (Detection of elements)			
Element	**Sodium Extract (S.E.)**	**Confirmed Test**	**Reaction**
Nitrogen	$Na+C+N \xrightarrow{\Delta} NaCN$ (S.E.)	S.E. + $FeSO_4$ + NaOH, boil and cool, + $FeCl_3$ + conc. HCl Blue or green colour	$Feso_4 + 2NaOH \longrightarrow Fe(OH)_2 + Na_2SO_4$ $Fe(OH)_2 + 6NaOH \longrightarrow Na_4[Fe(CN)_6] + 2NaOH$ $Na_4[Fe(CN)_6] + FeCl_3 \xrightarrow{HCl} Fe_4[Fe(CN)_6]_3 + 3$ NaCl Prussian blue
Sulphur	$2Ns + S \xrightarrow{\Delta} Na_2S$ (S.E.)	(i) S.E. + sodium nitro prusside (A) A deep violet colour. (ii) S.E. + CH_3COOH + $(CH_3COO)_2Pb$ A black ppt.	(i) $Na_2S + Na_2[Fe(CN)_5NO] \longrightarrow Na_4[Fe(CN)_5NOS]$ Deep violet (ii) $Na_2S + (CH_3COO)_2Pb \xrightarrow{CH_3COOH} PhS\downarrow + 2CH_3COONa$ black ppt.
Halogen	$Na + Cl \xrightarrow{\Delta} NaCl$ (S.E.)	S.E. + $HNO_3 = AgNO_3$ (i) White ppt. so lub le in aq NH_3 confirms Cl (ii) Yellow ppt partially so lub le in aq. NH_3 confirm br (iii) Yellow ppt inso lub le in aq NH_3 confirms I	$NaX + AgNO_3 \xrightarrow{HNO_3} AgZ\downarrow$ $AgCl + 2NH_3(aq) \longrightarrow [Ag(NH_3)_2]Cl$ White ppt soluble
Nitrogen and Sulphur together	$Na + C + N + S \xrightarrow{\Delta} NaCNS$ (S.E.)	As in test for nitrogen; instead of greed or blue colour, blood red coloration confirms presence of N and S both	$NaCNS + FeCl_2 \longrightarrow [Fe(CNS)]Cl_2 + naCl$ Blood red colour

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Beilstein Test :

The Beilstein test is a simple qualitative chemical test for organic halides. A copper wire is cleaned and heated in a Bunsen burner flame to form a coating of copper(II) oxide. It is then dipped in the sample to be tested and once again heated in a flame. A positive test is indicated by a green flame caused by the formation of a copper halide. The test does not detect fluorine/fluorides.

Quantitative Analysis of Organic Compounds (Kjeldahl's method, Dumas method, Carius method):

Estimation of carbon and hydrogen:

Both carbon and hydrogen are estimated in one experiment. A known mass of an organic compound is burnt in the presence of excess of oxygen and copper (II) oxide. Carbon and hydrogen in the compound are oxidised to carbon dioxide and water, respectively.

$C_xH_y + (x + y/4)\ O_2 \longrightarrow x\ CO_2 + (y/2)\ H_2O$

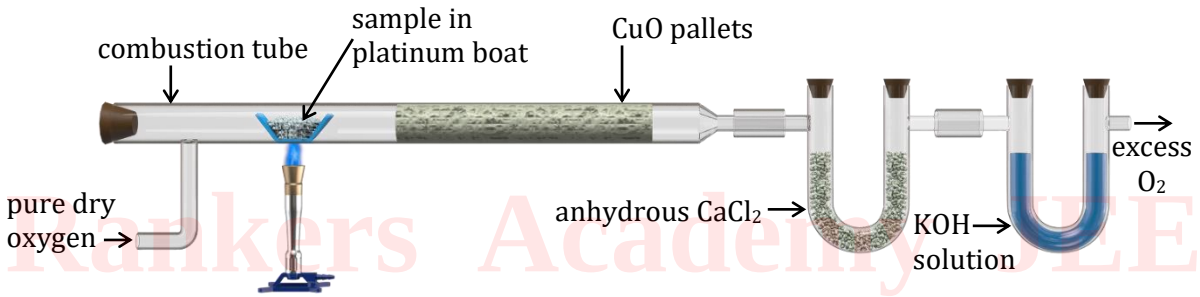


Figure : Estimation of carbon and hydrogen.
Water and carbon dioxide formed on oxidation of substance are absorbed in anhydrous calcium chloride and potassium hydroxide solutions respectively contained in U tubes.

The mass of water produced is determined by passing the mixture through a weighed U-tube containing anhydrous calcium chloride. Carbon dioxide is absorbed in another U-tube containing concentrated solution of potassium hydroxide. These tubes are connected in series (Fig. 8.14). The increase in masses of calcium chloride and potassium hydroxide gives the amounts of water and carbon dioxide from which the percentages of carbon and hydrogen are calculated.

Let

The mass of organic compound = w g

Mass increase in tube $CaCl_2$ = ag = mass of H_2O

Mass increase in tube KOH = bg = mass of CO_2 gas

Percentage of hydrogen % H = $\frac{2}{18} \times \frac{a}{w} \times 100$

Percentage of carbon % C = $\frac{12}{44} \times \frac{b}{w} \times 100$

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Estimation of Nitrogen:

There are two methods for estimation of nitrogen:

(i) Dumas method (ii) Kjeldahl's method.

(i) Dumas method: The nitrogen containing organic compound, when heated with copper oxide in an atmosphere of carbon dioxide, yields free nitrogen in addition to carbon dioxide and water.

$C_xH_yN_z + (2x + y/2)\ CuO \longrightarrow x\ CO_2 + y/2\ H_2O + z/2\ N_2 + (2x + y/2)\ Cu$

Traces of nitrogen oxides formed, if any, are reduced to nitrogen by passing the gaseous mixture over a heated copper gauze. The mixture of gases so produced is collected over an aqueous solution of potassium hydroxide which absorbs carbon dioxide. Nitrogen is collected in the upper part of the graduated tube.

Let the mass of organic compound = m g

Volume of nitrogen collected = V_1 mL

Room temperature = T_1K

Volume of nitrogen at STP = $\frac{P_1V_1 \times 273}{760 \times T_1}$ (Let it be V mL)

Where p_1 and V_1 are the pressure and volume of nitrogen, p_1 is different from the atmospheric pressure at which nitrogen gas is collected.

The value of p_1 is obtained by the relation p_1= Atmospheric pressure – Aqueous tension

22400 mL N_2 at STP weighs 28 g.

V mL N_2 at STP weighs = $\frac{28 \times V}{22400}$ g ; Percentage of nitrogen = $\frac{28 \times V \times 100}{22400 \times m}$

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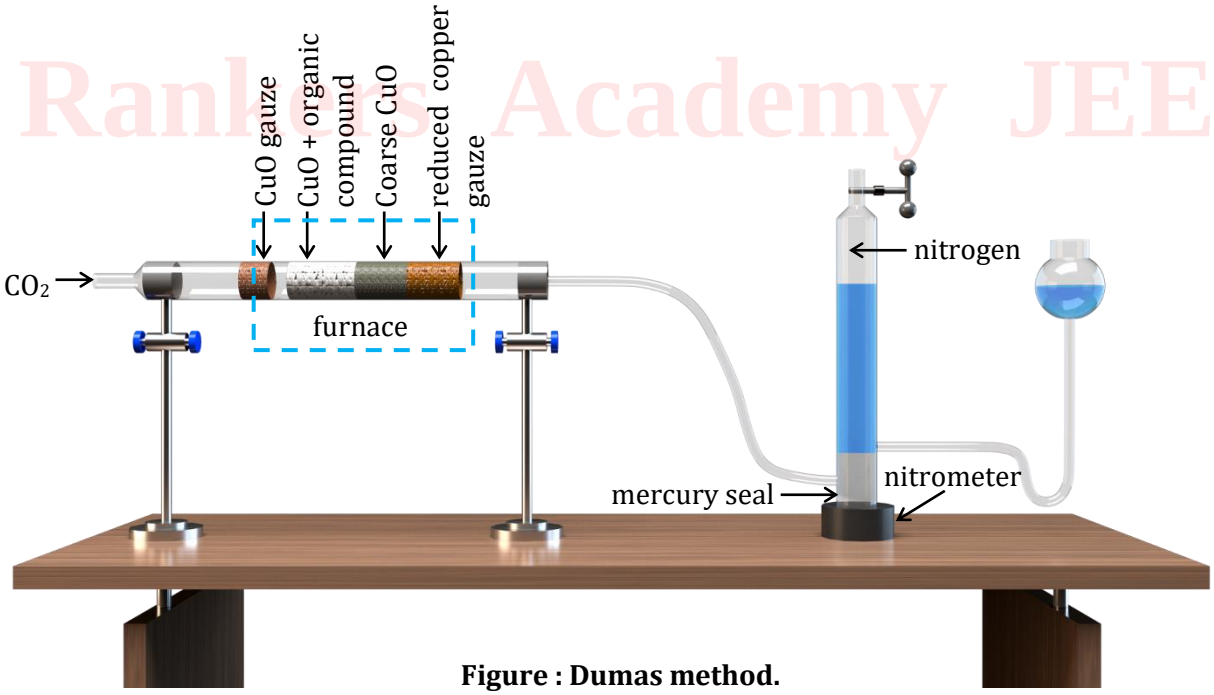


Figure : Dumas method.

The organic compound yields nitrogen gas on heating it with copper (II) oxide in the presence of CO_2 gas. The mixture of gases is collected over potassium hydroxide solution in which CO_2 is absorbed and volume of nitrogen gas is determined.

(ii) Kjeldahl's method:

$$\underset{\text{org. comp.}}{N} + \text{Conc. } H_2SO_4 \xrightarrow{CuSO_4,\ \text{Catalyst}} (NH_4)_2SO_4$$

$$(NH_4)_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O + 2NH_3$$

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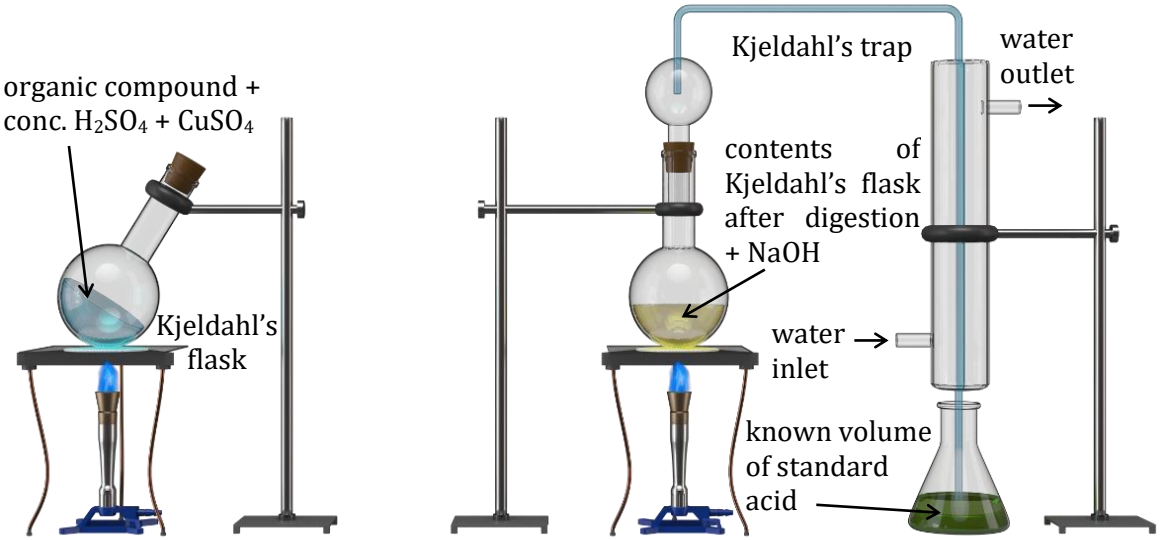


Figure : Kjeldahl method.

Nitrogen-containing compound is treated with concentrated H_2SO_4 to get ammonium sulphate which liberates ammonia on treating with NaOH; ammonia is absorbed in known volume of standard acid.

The liberated ammonia gas is absorbed in an excess of standard solution of sulphuric acid.

$$2NH_3 + \underset{\text{Left some acid}}{(H_2SO_4)} \longrightarrow (NH_4)_2SO_4$$

$$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$$

$$\% N = \frac{1.4 \times \text{meq. of } H_2SO_4 \text{ used to Neutralized } NH_3}{\text{mass of organic comp.}}$$

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NOTE : Kjeldahl method is not applicable to compounds containing nitrogen in nitro and azo groups and nitrogen present in the ring (e.g. pyridine) as nitrogen of these compounds does not change to ammonium sulphate under these conditions

Estimation of Halogens:

Carius method: A known mass of an organic compound is heated with fuming nitric acid in the presence of silver nitrate contained in a hard glass tube known as Carius tube.

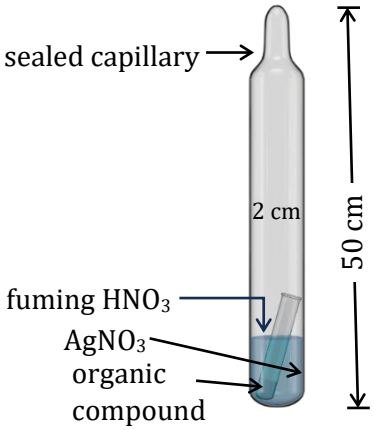


Figure : Carius method.

Halogen containing organic compound is heated with fuming nitric acid in the presence of silver nitrate.

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In a furnace, Carbon and hydrogen present in the compound are oxidised to carbon dioxide and water. The halogen present forms the corresponding silver halide (AgX). It is filtered, washed, dried, and weighed

$$\% \text{ of } X = \frac{\text{Atomic mass } X}{\text{Molecular mass AgX}} \times \frac{\text{mass of AgX}}{\text{mass of organic compound}} \times 100$$

$$\% \text{ of Cl} = \frac{35.5}{143.5} \times \frac{M_{AgCl}}{\text{mass of organic compound}} \times 100$$

$$\% \text{ of Br} = \frac{80}{188} \times \frac{M_{AgBr}}{\text{mass of organic compound}} \times 100$$

$$\% \text{ of I} = \frac{127}{235} \times \frac{\text{mass of AgI}}{\text{mass of organic compound}} \times 100$$

Estimation of Phosphorus:

$$\underset{\text{org. comp.}}{P} + HNO_3 + H_2O \longrightarrow H_3PO_4$$

$$H_3PO_4 + MgCl_2 + NH_4Cl + NH_3 \longrightarrow MgNH_4PO_4 \longrightarrow Mg_2P_2O_7$$

Let the mass of organic compound taken= W g

mass of $Mg_2P_2O_7$ formed = m g

$$\% \text{ P} = \frac{62 \times m}{222 \times W} \times 100$$

It is precipitated as ammonium phosphomolybdate, $(NH_4)_3PO_4.12MoO_3$, by adding ammonia and ammonium molybdate

The mass of organic compound taken = W g

mass of ammonium phosphomolybdate = mg

$$\text{Percentage of phosphorus} = \frac{31 \times m \times 100}{1877 \times W}$$

Estimation of Oxygen:

$$\text{Organic compound} \xrightarrow{\Delta} O_2 + \text{other gaseous products}$$

$$2C + O_2 \xrightarrow{1373\ K} 2CO \times 5$$

$$I_2O_5 + 5CO \longrightarrow I_2 + 5CO_2 \times 2$$

$$2I_2O_5 + 10C + 5O_2 \longrightarrow 2I_2 + 10CO_2$$

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Let

The mass of organic compound taken = m g

Mass of carbon dioxide produced = m_1g

$\because$ 88 g CO_2 is obtained if 32 g oxygen is liberated

$\therefore$ m_1g carbon dioxide is obtained from $\frac{32\ m_1}{88}$ g O_2

$\therefore$ Percentage of oxygen = $\frac{32\ m_1 \times 100}{88\ m}$ %

The percentage of oxygen can be derived from the amount of iodine produced also.

Estimation of Sulphur:

$$\underset{\text{org.comp.}}{S} + H_2O + \underset{\text{peroxide}}{3O} \longrightarrow H_2SO_4$$

$$H_2SO_4 + \underset{\text{excess}}{BaCl_2} \longrightarrow \underset{\text{white PPt}}{BaSO_4} + 2HCl$$

Let, The mass of organic compound taken = W g

mass of barium sulphate formed = m g

$$\% \text{ of } S = \frac{32}{233} \times \frac{m_{BaSO_4}}{W_{org.}} \times 100$$

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